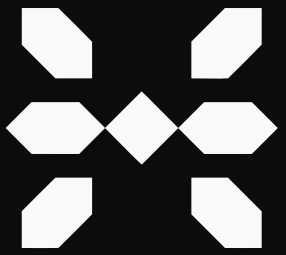
HOOX · PYNE · DOCUMENTATION



RUNTIME ENGINEERS · STRATEGY AUTHORS

RUNTIME MANUAL

Evaluator, series model, builtins, strategy, compiler

Open-source Pine Script evaluation framework

2026-10-17 DIN A4 · HTTPS://HOOX.SH/PYNE/DOCS OPEN SOURCE

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OVERVIEW

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title: "Runtime" description: "Bar-loop evaluator — series caps, incremental TA, alerts, fill export, foreign request.security → na, and interpret↔compile parity."

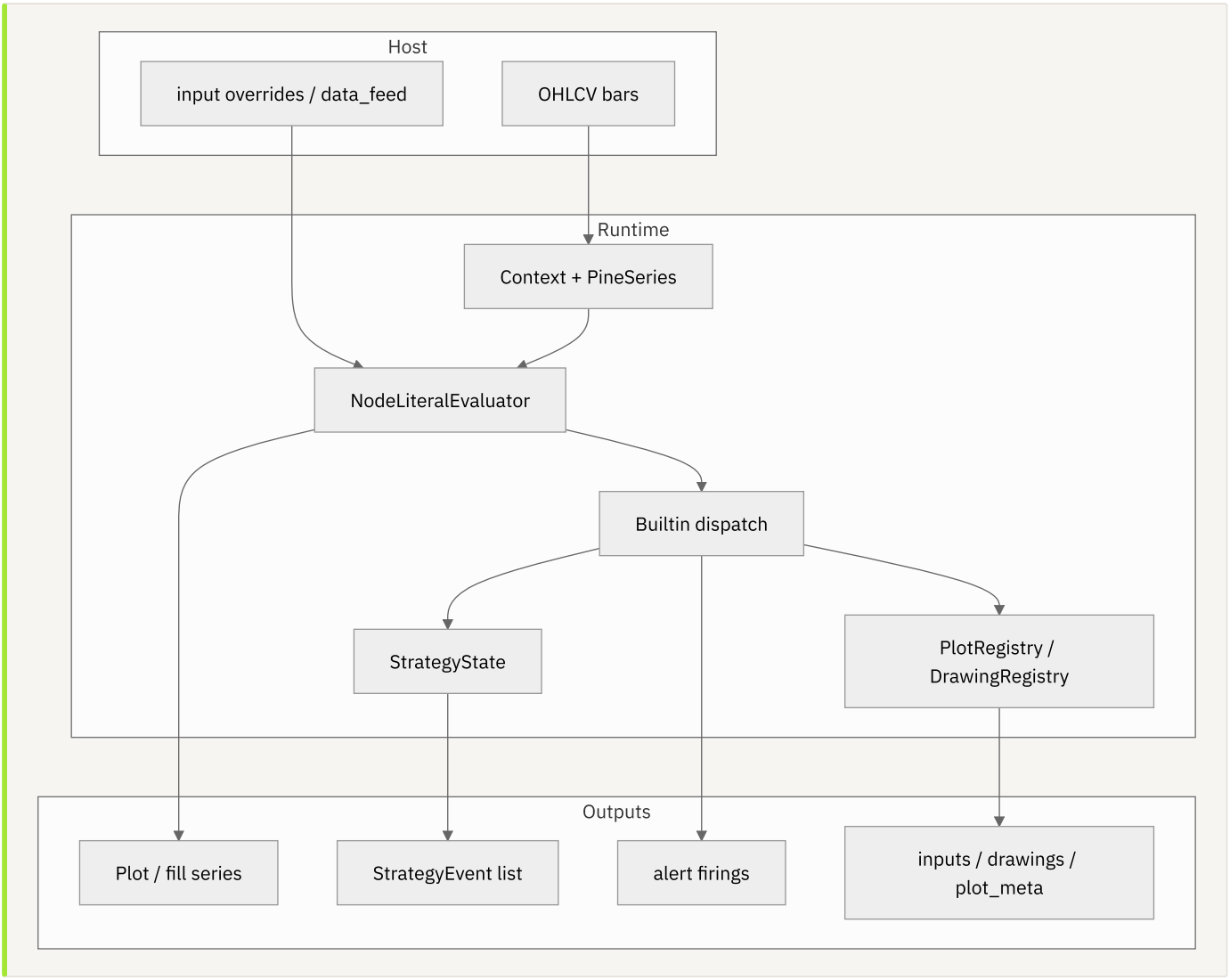
RUNTIME

Abstract

PYNE's runtime is the deterministic **bar-loop** that turns a parsed ASDL AST into series values, drawing side-effects, strategy events, and **alerts**. It is deliberately host-agnostic: the same visitor mixins power the library API, the Flask Pro API (`Runtime.run`), browser Pyodide, and edge workers. A second path—source-to-source compilation into a Numba or object-mode bar loop—trades interpretive flexibility for throughput on long histories; Pro API defaults to `mode=auto`.

This section documents **semantics**, not the chart. AXIS consumes plots / `fill` / drawings; hosts export alerts (optional L2 webhooks); HOOX consumes strategy events. None of those hosts is required to evaluate a script.

Conceptual model



Execution modes share the same parse tree (Pro API / `Runtime` also expose `auto`):

Mode	Entry	Bar loop	Typical use
Interpret	`NodeLiteralEvaluator.visit(tree)` per bar	Host updates `PineSeries` / context, re-visits AST	Full language surface, strategy, drawings, alerts, `request.*`
Compile	`pynescript.compiler.engine.compile_script`	Generated `for __bar_idx in range(n)` over OHLCV + **`time_arr`**	Numeric indicators (Numba) or object-mode subset
Auto	`Runtime.run(..., mode="auto")`	Compile when safe; interpret on fallback	Pro API default — prefer throughput without hard-failing exotic scripts

Compile inputs are contiguous arrays: `open_arr` … `vol_arr` plus **`time_arr`** (bar-open Unix ms; synthetic `bar_index * 60_000` when the host omits times) so `time` / calendar builtins match the interpret `time` series. Disk IR cache + product warm-compile (prewarm API/CLI) cut cold JIT latency. See Compiler overview and Interpret ↔ compile parity.

`request.security` (landed): interpret resamples same-symbol simple OHLCV and allowlisted `ta.sma` / `ema` / `rsi` / `atr` on a **coarser** TF (last **completed** HTF bucket only). Compile passthrough is same-symbol simple OHLCV only. Foreign symbols and complex / nested expressions resolve to **`na`** on both hosts (no inventing chart close as foreign fundamentals). Policy tags land on `result["meta"]["request_security"]`. Real multi-symbol feeds remain adapter work (roadmap **B1**).

Series caps / incremental TA: host list trim defaults **on** (`PYNE_SERIES_CAP`); bar-mode TA hot path defaults **on** (`PYNE_TA_INCREMENTAL`). Drawing GC honors `max_*_count` on the interpret registry; `fill` exports series keys + `plot_meta` for AXIS bands.

Interpret hot path (0.3.10 / Round 9): the AST walker inlines Assign / `Expr(Call)` after the first bar (function/type/import decls stay locked), skips unused derived OHLCV series (`hl2` / `hlc3` / `ohlc4` / `tr` unless named, `input.source`, or `ta.vwap`), and reuses plot cells by call-site. Same-machine bench @ 2000 bars vs 0.3.9: minimal **2.7×**, `ta.sma` **2.0×**, `ta_combo` **1.45×**. `PYNE_SERIES_RING` stays **off**.

Host flags and knobs

Knob	Default	Effect
`PYNE_SERIES_CAP`	**on**	Trim chronological `current_series` lists (`max_bars_back` / 256)
`PYNE_TA_INCREMENTAL`	**on**	Call-site incremental `ta.*` in bar mode
`PYNE_SERIES_RING`	**off**	Chronological ring lookback; skip dual list write when on
`PYNE_LIGHT_PLOTS`	**off**	Skip plot columns + input meta (corpus OK/fail only)
`PYNE_RUNTIME_MODE`	`interpret`	Default when `Runtime.run(mode=)` is omitted (`auto` is the Pro API body default)
`timeout_seconds=`	`None`	Interpret wall-clock budget; checked every 32 bars → `timed_out` + `error_kind=runtime`
`libraries=`	`[]`	`{namespace, name, version, source}` registered before `import` (auto forwards into interpret fallback)
`realtime_*`	historical	Interpret-only `varip` / forming-bar simulation (`realtime_last_bar` / `ticks` / `bars` / `from_bar`)
`Runtime.create_session`	interpret	Persistent `InterpretSession` (0.6.1): delta `append_bars`, forming-bar `update_last_bar`; compile has no resumable state

Interface surface

Concern	Start here
Series history, `[]`, `var` / `varip`, `PYNE_SERIES_CAP`, `na`	Series & history
Expressions, statements, control flow	Expressions & statements
Builtin namespaces (`ta.*`, `strategy.*`, ...)	Builtins hub
Drawing / `plot` / `fill` / `max_*_count` GC	Drawing & plotting
`request.*` / `input.*` (foreign → `na`)	request / input
Library `export` / `import`	Libraries
`StrategyEvent` parity contract	Events
`alert()` / webhooks	Alerts
Numba / object-mode compiler	Compiler overview
Interpret vs compile plot parity	Parity testing
Incremental interpret sessions (`create_session`)	Evaluate scripts

Library callers typically use `NodeLiteralEvaluator` or the package **`pynescript.runtime.Runtime`** (Pro API re-export) rather than wiring the bar loop by hand:

```
from pynescript.ast.evaluator import NodeLiteralEvaluator
from pynescript.runtime import Runtime  # package SoT (0.3.4+)

ev = NodeLiteralEvaluator(context={"close": [1.0, 2.0, 3.0], "bar_index": 2})
result = ev.evaluate_script('//@version=6\nindicator("x")\nplot(close)')

# Full bar loop (preferred host API):
# Runtime(symbol="CHART").run(
#     source, ohlcv_data=bars, mode="auto",
#     timeout_seconds=None, libraries=None,
# )
# Incremental interpret (0.6.1):
# sess = Runtime(symbol="CHART").create_session(source, bars)
# sess.append_bars(new_closed_bars)
# sess.update_last_bar(forming_bar)
```

Hosts that own OHLCV (Pro API, workers) use the package host: push bar → fill pending orders → `visit(tree)` → drain events → collect plots. **SoT:** `src/pynescript/runtime/` (`host.py`, `series.py`, `evaluator.py`). `backend.runtime` / `backend.series` / `backend.evaluator` are identity re-export shims for the Pro API.

Internals

Path	Role
`src/pynescript/runtime/`	Package Runtime SoT — host, series, CustomEvaluator
`src/pynescript/ast/evaluator/`	Mixin evaluators (base, names, expressions, statements, libraries, events)
`src/pynescript/ast/evaluator/builtins/`	Builtin dispatch tables
`src/pynescript/compiler/`	`CompilerVisitor`, Numba builtins, strategy broker, engine façade
`backend/runtime.py`	Compat re-export (`sys.modules` alias) of `pynescript.runtime.host`
`tests/test_evaluator.py`, `tests/test_parity.py`, `tests/test_strategy_*.py`, `tests/test_first_party_ta_goldens.py`	Semantic oracles + dual-host TA goldens

Composition of the full interpreter:

```
BaseEvaluator
  + LiteralEvaluator
  + NameEvaluator
  + ExpressionEvaluator
  + StatementEvaluator
  + BuiltinEvaluator
  → NodeLiteralEvaluator
```

`BuiltinEvaluator` itself aggregates mixins (`TechnicalAnalysisMixin`, `StrategyBuiltinsMixin`, `PlottingFunctionsMixin`, …) into one dispatch map built at construction time.

Invariants & edge cases

1. **Bar re-entrancy.** The AST is visited once per bar. Function/type definitions lock after bar 0 (`_pine_defs_locked`) so multi-dispatch tables do not grow (O(\text{bars}^2)).
2. **Order of broker steps.** Pending limit/stop fills run **before** script body evaluation on each bar (interpreter and compile object-mode agree). Pending-fill averaging when pyramiding ≤ 0 matches both brokers (**F2**).
3. **`na` is `None`.** Unresolved missing data, OOB history, and arithmetic with missing operands produce Python `None`, not IEEE NaN—except on the Numba path, which uses `np.nan` as the array sentinel. Cross-mode plot compares must normalize (`scripts/compare_interp_compile.py`).
4. **Series caps.** Unbounded `current_series` growth is capped (`PYNE_SERIES_CAP` default ON, `max_bars_back` / `_SERIES_MAX`) so long histories stay memory-bounded (**T1**).
5. **Incremental TA.** Hot-path kernels (MAs, oscillators, `bb` / `kama` / `cmo` / `stochrsi`, volume `obv` / `wad` / `cmf` / `klinger` / `nvi` / `pvi`, …) use call-site incremental updates in bar mode (`PYNE_TA_INCREMENTAL=0` to disable).
6. **Strategy state is per-evaluator.** `StrategyState` lives on the instance (`evaluator._strategy_state`), never as a process-global singleton—required for concurrent runs and parity tests.
7. **Drawing/plot registries are side channels.** Visual objects do not participate in pure expression values except where `plot()` returns a plot id for `fill()`. Hosts export `fill` bands for AXIS via series + `plot_meta` (and

`__drawings` on compile). Drawing objects honor `max_*_count` GC (package + Pro API + AXIS Pyodide).

8. **Alerts are a host side-channel.** `alert()` / `alertcondition()` firings are not strategy events; dual-host export + L2 webhooks are documented under Alerts.
9. **Host series are not aliased on assign.** `last = time` (or `open` / `high` /…) copies the current scalar into a fresh series for `last` —see Series & history.
10. **Omitted bid/ask stay `na`.** Host `bid` / `ask` update only when those keys appear on a bar dict; they are not synthesized from close.

Worked example — minimal bar loop mental model

```
for bar_index, bar in ohlcv:
    open/high/low/close.update(bar)
    context[bar_index, time, barstate, …] = …
    process_pending_orders(OHLC)   # broker sim
    evaluator.visit(ast)           # script body
    events += drain_events()
    series.append(plot values)
```

Compile mode collapses this into one generated function over full arrays (`open_arr` … `vol_arr`, `time_arr`), with `__bar_idx` standing in for `bar_index`.

Failure modes

Symptom	Likely cause
`unexpected type of node: …`	AST node not implemented in any mixin's `visit_*`
`Unknown built-in function: …`	Missing dispatch key or wrong qualified-name resolution
Divergent strategy events vs TV	Partial-fill / OCA / risk settings, or fill timing vs bar
Numba path raises / falls back	Script uses UDT/map/drawing → object mode; or Numba not installed
Plots empty but no error	Host forgot to collect `plot_outputs` / `PlotRegistry` after visit; or `PYNE_LIGHT_PLOTS=1`
`timed_out` + partial series	`timeout_seconds` elapsed (checked every 32 interpret bars)

See also

- Language core — grammar, AST, types before evaluation
- Alerts — `alert()` engine + L2 webhooks
- Pro API runtime bridge
- Interpret ↔ compile parity — `scripts/compare_interp_compile.py`
- Numerical validation
- Roadmap — H1/H2/T1/T2/F1/F2/L2/P1p official builtins landed; optional set0x unmeasured
- pyne-worker
- AXIS docs — optional AXIS for series/drawings

SERIES AND HISTORY

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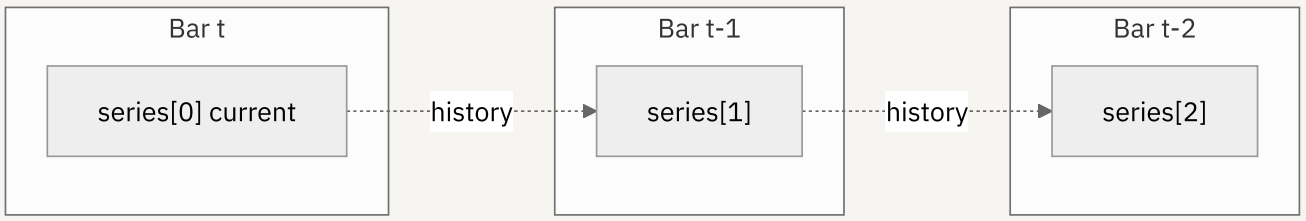
title: "Series and history" description: "Pine series model: history operator, na propagation, var/varip persistence, and bar-mode scalars."

SERIES AND HISTORY

Abstract

Pine's core data structure is the **series**: a value that evolves bar by bar, with random access to past values via the history operator `[]`. PYNE implements that model with host-managed series wrappers (`PineSeries` in the Pro API runtime), plain Python lists in unit tests, and flat `numpy` arrays on the compile path. Understanding history indexing and `na` is prerequisite to every builtin and strategy claim.

Conceptual model



Pine indices are **most-recent-first**:

Pine	Meaning	List representation (chronological)
`close[0]`	Current bar	`list[-1]`
`close[1]`	Previous bar	`list[-2]`
`close[n]` with (n \ge \text{len})	Out of range	`na` (`None`)

Negative offsets are **not** Python wraparound; they **soft-fail** to `na` (`None`).

Interface surface

History operator (`visit_Subscript`)

Implemented in `NameEvaluator.visit_Subscript` :

- Float indices coerce to `int` (e.g. `depth / 2`); NaN index → `na` .
- Lists map Pine index (i) to Python index `-(i + 1)` .
- Scalars: `x[0]` is `x` ; `x[i]` for (i > 0) is `na` (no history buffer).
- Matrices use `[row, col]` (tuple/list of length 2), not series semantics.

Series wrappers

Hosts expose OHLCV as objects with:

- `.current` — scalar for the active bar
- `.history` — most-recent-first buffer (deque or list)

Arithmetic in `ExpressionEvaluator` coerces such wrappers via `_as_scalar_operand` so `close + 1` does not attempt object addition.

Assigning host series (copy, not alias)

Binding a **host** multi-bar series into a **new** name copies the current scalar into a fresh series for that name—it does **not** alias the underlying `open` / `high` / `low` / `close` / `volume` / `time` buffers:

```
last = time          // copy of current time into series `last`
last := last + 1     // mutates only `last`, never the host `time` buffer
```

Implemented in `StatementEvaluator._bind_series_name` : if the RHS has `.current` / `.history` / `.update` (a host or tracked series), the RHS is reduced to its **current scalar** before allocating or updating the LHS series. Same-bar reassignment (`x = 0` then `x := expr`) overwrites the current sample so `x[1]` remains the previous bar's

final value. This prevents scripts such as dividend TTM time tracking from corrupting `time[j]` history when they rebind through a local.

UDF series parameters (0.6.2)

A Pine function parameter that is subscripted (`s[1]` inside `f(s) => …`) is a **series of that argument**, not a scalar that happens to be named `s` . Interpret now allocates a per-call-site history wrapper when the caller passed a scalar or a host-series current (`src = close; updown(src)`). Compile already tracked UDF series state; the two hosts agree on official TradingView® builtins (P1p **0 MISMATCH**). Nested calls get distinct streams. See Expressions & statements.

`var` / `varip` / `const`

Qualifier	Semantics in PYNE
`var` / `varip`	Initializer runs on first **execution** of that declaration site (tracked in `_var_declarations`), not only `bar_index == 0` . Later bars skip re-init so the value carries. History-tracked `var` series get a **start-of-bar carry** (`_commit_unwritten_history`) so `x[1]` is last bar's persist even when no `:=` ran this bar. On realtime ticks, `varip` re-evaluates the RHS.
`const` (v6)	Always initializes when the statement runs; not a cross-bar carry lock like `var` .
bare assign	Re-evaluates every bar; series "history" is the host series or prior list values.

Nested `var` inside `if barstate.islast` or a function therefore initializes on first path-taken bar—matching Pine's execution-based persistence.

`na` propagation

Operation	Rule
Binary arithmetic / comparison	Any `None` operand → `None` (or element-wise for lists)
Division by zero	→ `None` (not exception)
Soft type errors (`"a" + 1`)	→ `None`
Unary	`None` stays `None`
History OOB	→ `None`
Bare name `na`	Builtin / sentinel resolving to `None`
`nz(x, r)` (Numba path)	`nan` → replacement

List-valued series apply operators element-wise, aligning on the **trailing edge** when lengths differ (pad leading `None`).

`ta.highestbars` / `ta.lowestbars` offsets

These return a **bars-back offset** (not the extreme price):

Result	Meaning
`0`	Extreme is on the current bar
`-1`, `-2`, ...	Extreme is 1, 2, ... bars ago (down to `-(length-1)`)
`-1` sentinel (warm-up / invalid / all-`na`)	Short history (`bar_index+1 < length`), bad length, or no finite samples in the window

On ties, the **oldest** extreme in the window wins (leftmost bar). Interpret and Numba kernels share this contract (`_highestbars` / `numba_highestbars`).

History subscript interaction: Pine history indices are non-negative (`series[n]` with (n \ge 0)). Negative indices soft-fail to `na` on interpret series wrappers so warm-up / auto-step-`-1` / `highestbars` misuse do not abort the bar loop. On the compile path, dynamic offsets are coerced float→int (NaN→0) and the computed array index is clamped to `[0, n)` so expressions like `high[-ta.highestbars(...)]` (future-looking form) soft-fail to `nan` near series end rather than raising.

Bar mode vs full-series mode

Technical helpers (`technical_submodules/core.py`) distinguish:

- **Full-series mode** (unit tests with explicit lists): `ta.sma` may return a full list of values.
- **Bar mode** (`_pine_bar_mode`): returns the **current scalar** so expressions like `ta.ema(close,12) - ta.ema(close,26)` stay numeric per bar.

History buffers for indicators are truncated to a rolling window (`_SERIES_MAX = 256` for wrapper histories) to avoid (O(n^2)) full-history recomputation every bar.

Host series caps (`PYNE_SERIES_CAP`)

Pro API / `pynescript.runtime.Runtime` optionally **trims** chronological `current_series` lists and related host history so long charts do not grow (O(\text{bars})) memory per series.

Knob	Default	Meaning
`PYNE_SERIES_CAP`	**on**	Enable trimming. Disable with `0` / `false` / `no` / `off` (oracle / debug only).
Cap size	`256` (`DEFAULT_SERIES_MAX`)	Raised when the script declares a larger `max_bars_back=…`.
`PYNE_SERIES_MAX`	unset	Absolute override of the cap (positive int).
PineSeries history floor	`1000`	Separate from list caps; raised by `max_bars_back` / `PYNE_SERIES_MAX`.

Correctness notes (see `src/pynescript/runtime/series.py`; `backend/series.py` is a re-export shim):

- Window kernels (`ta.sma`, `highest`, ...) need `length ≤ cap`.
- Recursive smoothers under **incremental** TA (default `PYNE_TA_INCREMENTAL` on) carry state and stay safe independent of list length once warm.
- With **full recompute** (`PYNE_TA_INCREMENTAL=0`) and `bars ≫ cap`, EMA/RMA-style paths can diverge from a full-history oracle—prefer incremental (default) or raise the cap / disable series cap for goldens.

Out-of-range history offsets still return `na` (`None`); never `0`.

Unused derived series (0.3.10)

The interpret host updates `open` / `high` / `low` / `close` / `volume` / `time` every bar. Derived series (`hl2`, `hlc3`, `ohlc4`, `tr`, `time_close`) are written only when the source names them, uses `input.source`, or calls **`ta.vwap`** / **`vwap`** (default source is `hlc3` even when that identifier is absent). `ta.ao` rebuilds `hl2` from high/low if the derived list is empty.

`PYNE_SERIES_RING` (chronological tail view) remains **default off**. When on, the host skips the second chronological list write.

Internals

Path	Role
`src/pynescript/ast/evaluator/names.py`	`visit_Name`, `visit_Attribute`, `visit_Subscript`
`src/pynescript/ast/evaluator/expressions.py`	NA-safe binary/unary ops, series coerce
`src/pynescript/ast/evaluator/statements.py`	`var` / `varip` / `const` assign
`src/pynescript/ast/evaluator/builtins/technical_submodules/core.py`	`_as_series`, bar mode finalize
`src/pynescript/ast/evaluator/builtins/utility.py`	`max_bars_back`, `last_bar_index`, `na` helpers
`src/pynescript/runtime/host.py`	Bar loop; derived-series skip; series-cap trim (`backend.runtime` re-exports)
`src/pynescript/runtime/series.py`	`PineSeries`, `resolve_series_cap`, `PYNE_SERIES_CAP` / `PYNE_SERIES_MAX` / `PYNE_SERIES_RING`

Compile path: series are `np.full(n_bars, np.nan)` written at `__bar_idx`; history is `arr[__bar_idx - n]` with OOB → `np.nan` (see Compiler overview). Host **`time_arr`** is a chronological `float64` vector of bar-open Unix ms (same length as OHLCV); bare `time` lowers to `time_arr[__bar_idx]`, and `time[n]` uses the same history indexing rules as `close[n]`. When the host omits bar times, `Runtime` synthesizes `i * 60_000` so length always matches—calendar / `time` scripts stay defined on both modes.

Invariants & edge cases

1. **Chronology of `.history`.** Most-recent-first; reverse when converting to chronological lists for `ta.*` / `array.*`. Compile arrays are chronological (index 0 = first bar).
2. **No negative Pine indices.** Soft-fail to `na` / `nan` (not Python wraparound, and **not** a raise)—different from Python lists. See highestbars note above.
3. **`None` vs `float('nan')`.** Interpreter prefers `None`; Numba kernels use `np.nan`. Cross-mode comparisons must normalize.
4. **`max_bars_back`.** Raises the host series cap when larger than the default 256; still not a TV-style hard chart depth, but it affects trim policy.
5. **`PYNE_SERIES_CAP` default on.** Disable only for debugging / full-history oracles; pair with incremental TA for production long histories.
6. **Derived skip.** `ta.vwap()` still requires a live `hlc3` series; the host treats `vwap` as a derived-series consumer.

7. **Tuple returns from multi-value `ta.*`.** Unpacking `[a, b, c] = ta.macd(...)` uses `current` when it is a sequence of matching arity; otherwise history heuristics apply.
8. **Host series assign is by value.** New names never share the host OHLCV/`time` buffer reference.

Worked examples

History lag

```
//@version=6
indicator("lag")
delta = close - close[1]
plot(delta)
```

On bar 0, `close[1]` is `na` → `delta` is `na`. On bar 1+, arithmetic is ordinary floats (or series lists in batch tests).

`var` counter

```
//@version=6
indicator("count")
var int n = 0
n := n + 1
plot(n)
```

`n` initializes once; subsequent bars reassignment (`:=`) increments. Declaration sites are recorded in `_var_declarations`.

Element-wise NA

Given two list series of different lengths, `a + b` aligns tails and pads the longer prefix with `None`—preserving "most recent bars line up" intuition.

Failure modes

Symptom	Cause
All `na` after first bar	Host not updating series / `bar_index` between visits
`var` re-inits every bar	Fresh evaluator or `reset_var_declarations` each bar incorrectly
TypeError on `close + 1`	Series wrapper missing `.current` / not coerced
Compile vs interpret plot mismatch	`nan` vs `None`, or seed differences in EMA/RSI kernels — see parity
`time[j]` wrong after rebinding `last = time`	Expected only if host series were aliased; verify copy-on-assign path
`highestbars` always `-1`	Warm-up (`bar_index+1 < length`) or all-`na` window
Long-chart OOM / growing RSS	Series cap disabled or `PYNE_SERIES_MAX` huge; leave `PYNE_SERIES_CAP` on
EMA drift after many bars with cap	Full recompute + short cap — enable incremental TA or raise cap

See also

- Expressions & statements
- Technical builtins
- Numba path
- Interpret ↔ compile parity
- Runtime hub

EXPRESSIONS STATEMENTS

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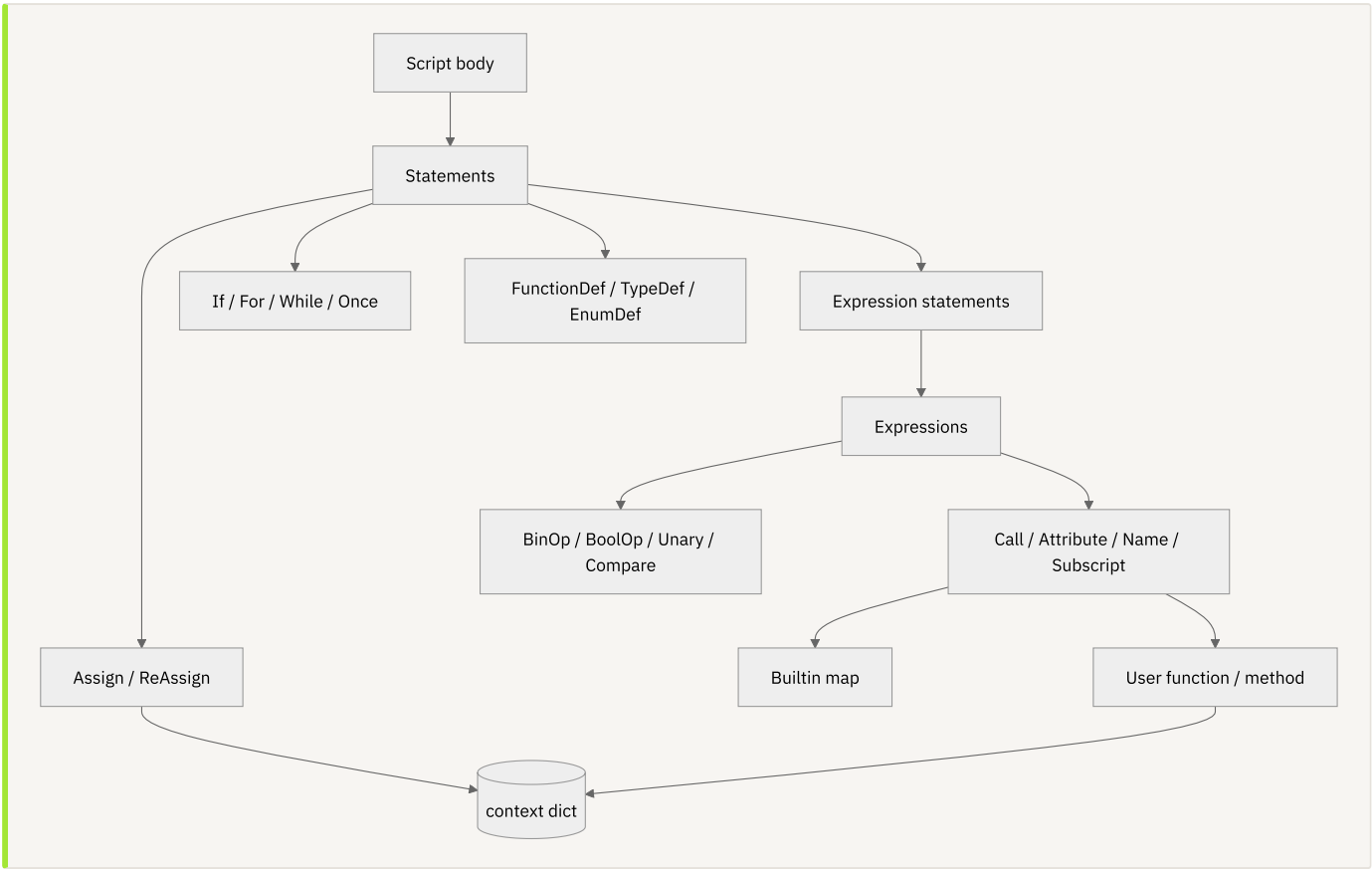
title: "Expressions and statements" description: "How the AST walker evaluates operations, calls, control flow, assignments, and user definitions."

EXPRESSIONS AND STATEMENTS

Abstract

Evaluation is pure visitor dispatch: each ASDL node type maps to a `visit_*` method on a mixin. Expressions produce values (with Pine `na` and series rules); statements mutate context, register types/functions, or emit side effects via builtins. This page is the operational semantics of the interpreter path—what runs inside `evaluator.visit(tree)` each bar.

Conceptual model



Interface surface

Expressions (`ExpressionEvaluator`)

Node	Behavior
`BoolOp` (`and` / `or`)	Manual short-circuit: `and` stops at first falsy (`na` / `0` / `False`); `or` stops at first truthy. Returns a Python `bool` .
`BinOp`	NA-safe `+ - * / %` ; division by zero → `na`
`UnaryOp`	`not` , unary `+` / `-` with NA and list map
`Compare`	Chained comparisons with NA-safe operators
`IfExp`	Ternary: condition, then, else
`Call`	Builtin dispatch, user functions, methods, multi-dispatch overloads
`Tuple` / lists	Literal sequences for unpacking and multi-arg returns

Binary ops coerce `PineSeries` -like objects to `.current` before operating. List operands broadcast or zip with trailing alignment (see Series & history).

Names (`NameEvaluator`)

Node	Behavior
`Name`	Context lookup; bare series builtins (`na` , `year` , ...); else string id (lazy)
`Attribute`	Qualified context keys, builtins (`strategy.position_size`), library modules, UDT fields/methods, array/drawing/matrix method markers, Python getattr fallback
`Subscript`	Series history / array / matrix indexing

Critical: qualified names like `strategy.opentrades.entry_price` are resolved via AST path construction (`ast_qualified_name`) **without** evaluating intermediate zero-arg series that would collapse the path to an int.

Statements (`StatementEvaluator`)

Construct	Behavior
`Script`	Walk body; finalize library exports if `library(...)` active
`Assign`	`var` / `varip` once-per-declaration; `const` ; ordinary assign; tuple unpack; `export` registration. History-tracked `var` series get a start-of-bar carry so `x[1]` is the prior persist.
`ReAssign` (`:=`)	Update existing binding (and UDT fields)
`If` / `For` / `While`	Standard control flow; loop variables in context
`Once`	Fire body the first time `test` is true (omit `test` → `true`) on a **confirmed** bar, then skip. Unconfirmed realtime ticks may re-run until `barstate.isconfirmed` commits the fired flag. Does not return a value.
`FunctionDef`	Store callable in context; multi-dispatch overload lists; `export` ; methods tagged `__pine_method__` . After bar 0 the host walks a **hot body** that skips Function/Type/Enum/Import (already bound).
`TypeDef` / `EnumDef`	Type registry / enum member dicts
`Import`	Resolve `LibraryRegistry` → bind alias to `LibraryModule`
Declarations	`indicator` / `strategy` / `library` via declaration builtins

Function invocation binds parameters into context (with restore of prior bindings on exit) so nested calls and bar-local state interact cleanly. After bar 0, hosts set `_pine_defs_locked` so re-visiting the script does not re-register overload tables. The locked walk inlines `Assign` and `Expr(Call)` (no generic `visit` frame) and UDF/call-expression history keys use `_pine_site_id` stamped on the AST.

Calls and multi-dispatch

Method markers returned from attribute evaluation:

Marker	Meaning
`("_method_call", instance, name)`	UDT method
`("_array_method", list, name)`	`array.<name>(self, …)`
`("_ns_method", obj, "label.get_text")`	Drawing/matrix namespace method
`("_ext_method", receiver, name)`	Free `method` with receiver as first arg

Overloads with typed first parameters (e.g. `matrix.float` vs `array.label`) select via coarse type tags; `na` receivers deliberately exclude matrix/array/drawing tags so optional UDT fields do not mis-dispatch to `tostring` on collections.

Internals

Path	Role
`src/pynescript/ast/evaluator/expressions.py`	Ops, calls, conditionals
`src/pynescript/ast/evaluator/names.py`	Names, attributes, subscripts
`src/pynescript/ast/evaluator/statements.py`	Script structure, assign, defs, import
`src/pynescript/ast/evaluator/literals.py`	Constants, colors, strings
`src/pynescript/ast/evaluator/base.py`	Context, math constants, library registry hook
`src/pynescript/ast/evaluator/types.py`	`EvaluatorProtocol` typing for mixins

Invariants & edge cases

1. **Context is the single store.** Builtins read OHLCV and strategy series from `self.context`; hosts must keep it coherent each bar.
2. **String fallback names.** An unbound `Name` returns its id string so later attribute/call resolution can still hit the builtin map (`"ta.sma"` patterns via attribute chains).
3. **Parameter unbind.** Missing-before-bind params use a sentinel so unbind `pop`s rather than leaving `None` ghosts.
4. **`na` normalization.** String `"na"` / `"nan"` / `"none"` may coerce to `None` at dispatch boundaries for UDT optional args.
5. **User functions re-enter the visitor.** Recursive and higher-order patterns are limited by Python stack, not a Pine-specific trampoline.

Worked examples

Ternary and comparison chain

```
//@version=6
indicator("cmp")
v = close > open ? close - open : open - close
plot(v)
```

Each operator path is NA-safe: if `close` or `open` were missing, comparisons and arithmetic yield `na`.

User function with series

```
//@version=6
indicator("fn")
f(x, n) =>
    ta.sma(x, n)
plot(f(close, 14))
```

`f` is stored as a Python callable that rebinds `x` / `n` and visits the function body AST each call.

UDF series parameters (0.6.2): when a caller passes a scalar (or a host-series current) into a function that then reads `s[1]`, interpret wraps that parameter as a per-call-site history stream. `src = close; updown(src)` therefore matches compile — the previous-bar sample is the previous argument at that call site, not an empty scalar. Official TradingView® builtin scripts are 0 MISMATCH vs compile on this contract.

`once` (August 2026)

```
//@version=6
indicator("once")
var int n = 0
once close > ta.sma(close, 20)
    n := 1
plot(n)
```

Equivalent to a `var bool` latch plus `if`, but cheaper: the structure is not re-evaluated after it fires on a closed bar. On a forming realtime bar (`Runtime.run(..., realtime_last_bar=True, realtime_ticks=N)`), the latch is not committed until `barstate.isconfirmed`. Compile is historical (`isconfirmed` always true).

Intrabar rollback on realtime ticks (0.6.0)

On a forming realtime bar, intermediate ticks snapshot the `var` scope (context bindings, series currents, declaration set) and restore it after the visit — every tick starts from confirmed state and only the final tick commits. `varip` names persist across ticks by design. Shallow by design: history pushes, the series-assign map, in-place container mutations, and broker side effects are not rolled back.

Method multi-dispatch sketch

```
method log(this, string s) => ...
method log(this, float x) => ...
```

Both register under the same name with `__pine_overloads__`; call sites pick by receiver/arg type tags.

Failure modes

Error	Meaning
`unexpected type of node`	Missing `visit_*` for an AST construct
`Unsupported binary operator`	Op class not in the dispatch table
History `series[-1]` is `na`	Negative Pine offsets **soft-fail** to `None` (not Python wraparound, not a raise)
Wrong overload / destroyed receiver	Historical `na` string path; now normalized—report residual mismatches as bugs
Stack overflow	Deep recursion in user functions

See also

- Series & history
- Builtins hub
- Libraries

- Type system (core)

OVERVIEW

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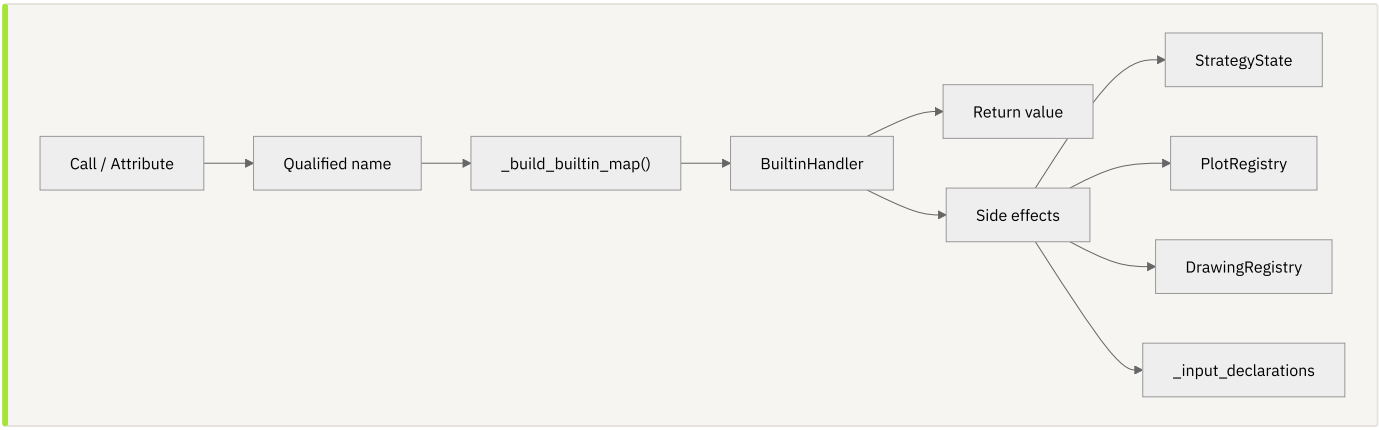
title: "Builtins" description: "Dispatch architecture for ta.*, strategy.*, collections, plotting, request.*, and other standard namespaces."

BUILTINS

Abstract

Pine's standard library is not a single module: it is a **flat qualified-name dispatch table** (`"ta.sma"`, `"strategy.entry"`, `"array.push"`, …) assembled from category mixins. Each entry is a handler `(args, kwargs?) → value` with side effects when the language requires them (orders, plots, drawings, inputs). This hub orients the surface; child pages go deep on technical, strategy, collections, drawing/plotting, and request/input.

Conceptual model



`BuiltinEvaluator` merges mixin maps in a fixed order and then registers ticker, logging, color, timeframe, and script-declaration functions. The `strategy()` declaration handler is wrapped so broker settings apply to `StrategyState`.

Interface surface

Namespace / area	Mixin / module	Docs
`ta.*`	`TechnicalAnalysisMixin` + `technical_submodules/*`	Technical
`strategy.*`	`StrategyBuiltinsMixin`, `StrategyConstantsMixin`	Strategy
`array.*` / `matrix.*` / `map.*`	`ArrayBuiltinsMixin`, matrix/map evaluators	Collections
`plot*` / `hline` / `line` / `label` / …	`PlottingFunctionsMixin`, `DrawingBuiltinsMixin`	Drawing & plotting
`request.*` / `input.*`	`RequestBuiltinsMixin`, `InputBuiltinsMixin`	Request & input
`math.*`, min/max, `nz`, …	`NumericBuiltinsMixin`	(this page)
`str.*`	`StringBuiltinsMixin`	(this page)
`color.*`	`register_color_functions`	(this page)
`syminfo` / `ticker.*`	`register_ticker_functions`	(this page)
`timeframe.*`	`register_timeframe_functions`	(this page)
`alert` / logging	`AlertsMixin`, `register_logging_functions`	Alerts
`indicator` / `strategy` / `library`	`register_script_declaration_functions`	Libraries
Footprint / volume_row	`FootprintBuiltinsMixin`	Request & input

Numeric and string essentials

- **Numeric:** `math.*` constants from `BaseEvaluator` (`math.pi`, `math.e`, golden-ratio helpers); functions cover abs, min/max, rounding, logs, trig, and NA helpers (`nz`, `na` predicates via utility).
- **String:** `str.*` formatting, conversion (`str.tostring` respects `format.*` constants), and manipulation aligned with common Pine scripts in the builtin corpus.

Color, ticker, timeframe

- **Colors:** hex constants (`color.red` → `#F23645`, etc.) plus composition helpers.

- **syminfo / ticker:** host-provided `Syminfo` object or flat context keys; ticker construction for `request.security` symbols.
- **timeframe:** period flags (`isintraday` , `isdaily` , ...) default daily in base constants; hosts override from bar spacing.

Alerts

`alert(message, freq)` and `alertcondition(condition, title, message)` record structured firings (`message` , `freq` , `bar_index` , `time` , `source`) with TV-style frequency rules (`once_per_bar` , `once_per_bar_close` , `all`). Hosts export them on `/run` and can POST last-bar batches to webhooks (**L2**). Side channel only — **not** strategy events.

Full detail: Alerts.

Internals

Path	Role
`src/pynescript/ast/evaluator/builtins/__init__.py`	`BuiltinEvaluator` composition
`src/pynescript/ast/evaluator/builtins/base.py`	Dispatch mixin, `_call_builtin` , registration
`src/pynescript/ast/evaluator/builtins/*.py`	Category implementations
`scripts/generate_builtin_metadata.py`	LSP/metadata surface (not hand-edited JSON)

Handlers are looked up by the fully qualified string built during name/attribute evaluation. Zero-arg series (e.g. `strategy.equity`) are registered as builtins that ignore empty args.

Invariants & edge cases

1. **One map, many mixins.** New builtins require a map entry *and* (usually) metadata regeneration for LSP.
2. **Kwargs and positionals.** Handlers accept both; Pine named arguments arrive as kwargs when the call AST carries them.
3. **Bar mode.** Stateful `ta.crossover` / similar use per-bar call indices reset by the host (`_cross_call_i`). Incremental `ta.*` (`PYNE_TA_INCREMENTAL` , default on) uses a sibling `_ta_call_i` slot.
4. **Mock fallbacks.** Bare equity-style `request.security` strings may still mock offline; foreign / fundamental prefixes and complex expressions return `na` (see request / input).
5. **Compile path subset.** Numba builtins reimplement only a fraction of this table; object mode covers more via Python helpers—see Compiler.

Worked example — resolution path

```
plot(ta.sma(close, 14))
```

1. Attribute/call chain builds qualified name `ta.sma` .
2. `_call_builtin("ta.sma", [close_series, 14])` .
3. Handler coerces series → list, computes SMA, finalizes scalar or list.
4. `plot` registers a `Plot` with that series value and returns the plot id.

Failure modes

Symptom	Fix
`Unknown built-in function`	Typo, version surface gap, or failed attribute chain
Always mock prices	Wire `data_feed` / `data_provider` into evaluator context
LSP knows a function runtime lacks	Metadata ahead of implementation—check missing features
Handler arity errors	Positional vs keyword mismatch in script

See also

- Technical
- Strategy
- Collections
- Drawing & plotting
- Request & input
- Alerts
- Builtin metadata (LSP)

TECHNICAL

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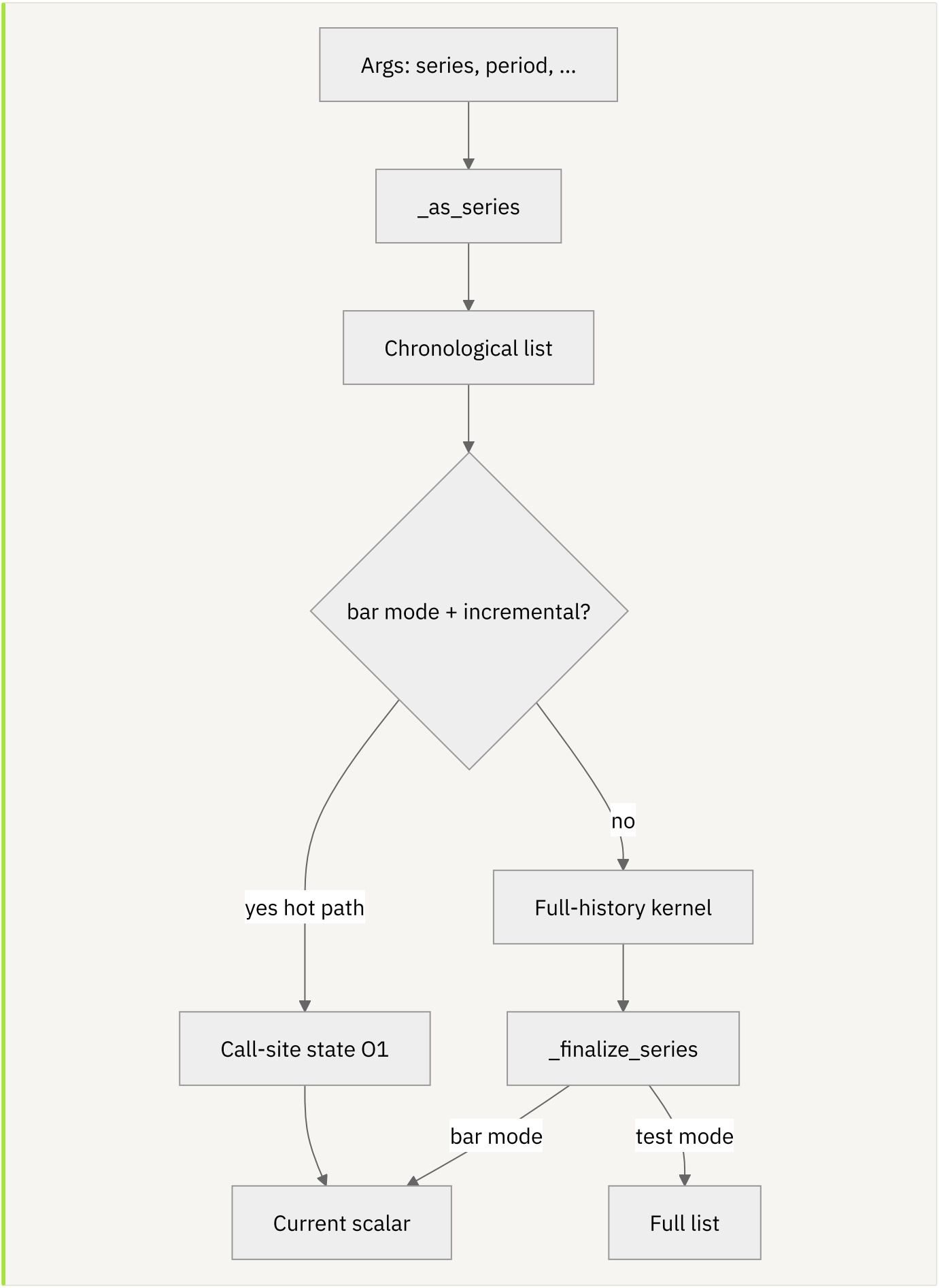
title: "Technical analysis (ta.*)" description: "ta.* indicators: series helpers, bar mode, interpret↔compile parity, and submodule layout."

TECHNICAL ANALYSIS (`ta.*`)

Abstract

The `ta.*` namespace is the largest pure-compute surface in PYNE: moving averages, oscillators, volatility bands, volume studies, pattern helpers, and cross/rise/fall predicates. Implementations live under `technical_submodules/` and are composed into `TechnicalAnalysisMixin`. Numerical behavior is validated against TradingView reference series (see numerical validation); the design goal is IEEE-limit parity, not “approximate TA.” Interpret and compile (Numba) hosts share the same formulas for the kernels listed under Interpret ↔ compile parity.

Conceptual model



Hosts in production set bar mode so indicator calls return **scalars for the active bar**, matching how Pine expressions compose (`ta.ema(a) - ta.ema(b)`).

Incremental hot path

When `_pine_bar_mode` and `_pine_ta_incremental` are enabled (Runtime default; disable with **`PYNE_TA_INCREMENTAL=0`**), hot `ta.*` builtins update **call-site state** once per bar instead of recomputing full history. Call sites are indexed like crossovers (`_ta_call_i` reset each bar). Nested forms such as `ta.ema(ta.sma(close, 14), 10)` stay correct because each call keeps its own slot. Golden suite: `tests/test_ta_incremental.py` (incremental last values ≡ full recompute).

Family	Incremental builtins (non-exhaustive)
Moving averages	`sma`, `ema`, `rma`, `wma`, `hma`, `vwma`, `swma`, `kama`, `dema`, `tema`, `alma`
Oscillators	`rsi`, `macd`, `stoch` / `stochrsi`, `cci`, `cmo`, `tsi`, `roc`, `wpr`, `aroon`, `dpo`, `kst`
Volatility / structure	`atr`, `tr`, `stdev`, `bb`, `highest` / `lowest`, `linreg`, `adx` / `dmi`, `supertrend`, `kc`, `sar`, `donchian`
Volume (**0.3.10–0.3.11**)	`obv`, `wad` / `wvad`, `cmf`, `klinger`, `mfi`, `vwap`, `accdist`, `nvi`, `pvi`
Other	`change` / `mom` / `cum`, `median`, `percentrank`, `rising` / `falling`, `barssince`, pivots

`ta.nvi` / `ta.pvi` are incremental (O(1)/bar; `PYNE_TA_INCREMENTAL=0` full-list). **`ta.atr` is Wilder RMA of true range** on interpret (full + incremental) and Numba (`ta.rma(ta.tr, length)` — 0.3.4+). Supertrend is locked mid±factor·ATR (not the TV band ratchet).

Interface surface

Dispatch keys (non-exhaustive; map is authoritative in `technical.py`):

Family	Examples
Moving averages	`ta.sma`, `ta.ema`, `ta.wma`, `ta.rma`, `ta.hma`, `ta.vwma`, `ta.swma`, `ta.alma`
Oscillators	`ta.rsi`, `ta.stoch`, `ta.cci`, `ta.cmo`, `ta.mfi`, `ta.roc`, `ta.wpr`, `ta.tsi`, `ta.rci`
Trend / channels	`ta.macd`, `ta.adx`, `ta.dmi`, `ta.supertrend`, `ta.sar`, `ta.linreg`
Volatility	`ta.atr`, `ta.tr`, `ta.bb`, `ta.bbw`, `ta.kc`, `ta.kcw`, `ta.stdev`, `ta.variance`
Volume	`ta.obv`, `ta.vwap`, `ta.mfi` (shared), volume submodule helpers
Structure	`ta.highest` / `lowest` / `highestbars` / `lowestbars`, `ta.pivothigh` / `pivotlow`
Predicates	`ta.crossover`, `ta.crossunder`, `ta.cross`, `ta.rising`, `ta.falling`
Stats / other	`ta.change`, `ta.mom`, `ta.cum`, `ta.median`, `ta.mode`, percentiles, `ta.barssince`, `ta.valuewhen`, `ta.correlation`

Argument conventions

- Typical form: `ta.fn(series, length)`.
- Some functions allow **period-only** calls (e.g. `ta.highest(20)`) defaulting source to `high` / context series via `_expect_series(..., allow_period_only=True)`.

- Multi-value returns (e.g. MACD, BB, Supertrend) unpack as tuples/lists; assignment uses StatementEvaluator unpack rules.
- Fractional lengths floor to int (TradingView-compatible).

Stateful crosses

`ta.crossover` / `ta.crossunder` need previous-bar pairs. In bar-mode runs the host resets a call-site index (`_cross_call_i`) each bar so multiple cross calls in one script keep independent state.

Internals

Path	Role
`src/pynescript/ast/evaluator/builtins/technical.py`	Dispatch map aggregation
`.../technical_submodules/core.py`	Series coerce, expect helpers, bar finalize, **incremental kernels**
`.../moving_averages.py`, `oscillators.py`, `volatility.py`, `volume.py`, …	Kernels + inc wiring
`.../common.py`, `basic.py`, `advanced.py`, `patterns.py`, `strategies.py`, `synthesizer.py`, `economics.py`	Additional families
`src/pynescript/compiler/numba_builtins.py`	Compile-path Numba kernels (must track interpret formulas)
`tests/test_ta_indicators_*.py`, `tests/test_indicators.py`	Regression
`tests/test_ta_incremental.py`	Inc ≡ full recompute golden (full CI gate)
`tests/test_first_party_ta_goldens.py`	Dual-host first-party goldens (ATR / Supertrend / Keltner)
`tests/test_interp_compile_parity.py`	Always-on dual-host smoke + full-corpus mark
`scripts/compare_interp_compile.py`	Corpus harness (report under `.cache/interp_compile_parity.json`)
`tests/test_compiler_numba.py` (`TestInterpCompilePlotParityFixes`, highestbars offsets, …)	Targeted formula locks
`docs/numerical_validation_report.md`	Published precision summary (TV / IEEE bounds)

History for wrapper series is reversed to chronological order and may be truncated (`_SERIES_MAX`) before full kernels run. Incremental path only needs `series[-1]` per call, so truncation does not freeze state.

Interpret ↔ compile parity

Numba kernels are **separate code** from the interpreter, but the dual-host contract is **same formula, same warm-up mask, same `na` rules** for the surface below. Drift is a bug, not an allowed “approx compile.” Prove regressions with the interp↔compile harness and the numerical validation page (TV / IEEE bounds).

Topic	Shared behavior
`ta.rsi`	**Wilder** on both hosts: SMA seed of the first `period` deltas, then RMA of gains/losses. First finite bar is index `period` (`period` deltas need `period+1` prices). Not a simple window average.
`ta.roc`	Standard TV formula `100 * (src - src[length]) / src[length]` . Warm-up is **`na`** , never an early `0.0` . Zero or missing baseline → `na` . Interpret no longer used a wrong lookback denominator or early zero.
`ta.wma`	Requires a **full non-`na` window** of length `N` . Nested forms such as `ta.wma(ta.roc(...), …)` stay `na` until the inner series has produced `N` finite samples (no partial-window reweight).
`ta.cum`	Running sum; Pine **`na` / IEEE NaN treated as 0** (skipped contribution), matching TradingView cumulative-sum semantics.
`ta.highestbars` / **`ta.lowestbars`**	Return **negative bars-back offsets**: `0` if the extreme is the current bar, `-1` one bar ago, …, `-(length-1)` at the far edge. Short history (`i + 1 < length`) returns `-1` . Matches Aroon-style scripts that index with `high[ta.highestbars(...)]` / `math.abs(ta.lowestbars(...))` .
`math.avg` **(multi-arg)**	Arithmetic mean of the arguments. **Any `na` argument** → **`na`** (do not skip). Not a rolling `ta.sma` .
`ta.linreg`	Least-squares fit over the window; **endpoint at `x = n−1` when `offset=0`** (TV): `mean_y + slope * ((n−1) − mean_x)` / compile `intercept + slope * (n − 1 − offset)` . Length < 2 → `na` .
`ta.mfi`	Warm-up aligned on both hosts: needs **`length + 1` typical-price samples** (direction vs previous bar). Equal typical prices contribute to neither side; only-pos / only-neg MF → 100 / 0 (or 50 when both empty).
`ta.rci`	Rank Correlation Index (Spearman of time vs value ranks) is implemented on the **compile path** (`numba_rci`) as well as interpret; stays in numeric mode when called.

0.3.4+: `ta.atr` is Wilder RMA of TR on interpret + Numba (not EMA-of-TR). EMA dual-host seed is full-list SMA seed matching incremental/Numba. First-party dual-host goldens (ATR / Supertrend / Keltner) and `test_ta_incremental` gate residual drift. Any remaining gaps are tracked under missing features—not as silent “close enough” for the rows above.

Parity harness

```
# Always-on smoke (stable scripts under tests/test_interp_compile_parity.py)
pytest tests/test_interp_compile_parity.py -q

# Corpus compare (default ~50 scripts × 1000 bars; writes report JSON)
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --glob 'average_*.pine' --bars 200

# Optional longer pytest path
pytest tests/test_interp_compile_parity.py -m interp_compile_full

# Formula locks (RSI/ROC/WMA/cum/math.avg/highestbars, …)
pytest tests/test_compiler_numba.py -k 'InterpCompilePlotParity or highestbars_lowestbars_negative'
```

Report path: `.cache/interp_compile_parity.json` . Methodology and acceptable relative-error bands vs TradingView: Numerical validation.

Invariants & edge cases

1. **Warm-up → `na`.** Insufficient bars (e.g. SMA length (N) needs (N) samples; RSI needs `period+1` prices) yield `None` / leading `na` in full-series mode.
2. **NA in window.** Many kernels propagate `na` if any window element is missing (SMA-like sums, **WMA**, MFI money-flow samples). **`ta.cum` is the TV exception** (`na` → 0 contribution).
3. **Default sources.** `ta.atr` pulls high/low/close from `current_series` / context when not passed explicitly.
4. **Compile kernels must track interpret.** Numba `numba_*` implementations are separate source; for the parity table above they are formula-locked by dual-host tests—do not "approximate" for speed.
5. **No chart look-ahead.** Kernels only see history available at the current bar index; `request.security` lookahead is a separate concern.
6. **Incremental ≡ full recompute (oracle).** ATR seed is already Wilder RMA on both hosts; changing seed rules is a correctness project, not a silent perf flag.

Worked examples

Classic overlay

```
//@version=6
indicator("SMA 14")
v = ta.sma(close, 14)
plot(v)
```

In bar mode each visit returns one float (or `na`); the host stacks them into a series for the response envelope.

Cross entry signal

```
//@version=6
strategy("X")
fast = ta.ema(close, 12)
slow = ta.ema(close, 26)
if ta.crossover(fast, slow)
    strategy.entry("L", strategy.long)
```

Cross state is bar-local; ensure the runtime resets cross call indices when reusing an evaluator.

Multi-value unpack

```
[macdLine, signal, hist] = ta.macd(close, 12, 26, 9)
plot(hist)
```

Failure modes

Symptom	Cause
Always `na`	Length longer than available history; or series not updated
Wrong vs TV by large margin	Wrong source series; mock OHLCV; or non-bar-mode list composition bug
Cross never fires	Call-index state not reset / shared incorrectly across bars
Compile diverges from interpret	Kernel drift in `numba_builtins.py` (RSI must stay Wilder; ROC warm-up must stay `na`; WMA full window; highestbars sign; `math.avg` na rule)—re-run the parity harness
Early ROC / nested WMA finite too soon	Legacy quirks: ROC returned `0.0` before lookback, or WMA reweighted partial/`na` windows—fixed on both hosts
`high[ta.highestbars(...)]` mostly `na`	Offsets are **negative**; only offset `0` indexes the current bar without `math.abs`
`math.avg` finite when an arg is `na`	Bug: multi-arg avg must propagate `na`, not skip

See also

- Series & history
- Strategy builtins
- Numba path
- Numerical validation
- Missing features (residual dual-host gaps)
- Compiler overview

STRATEGY

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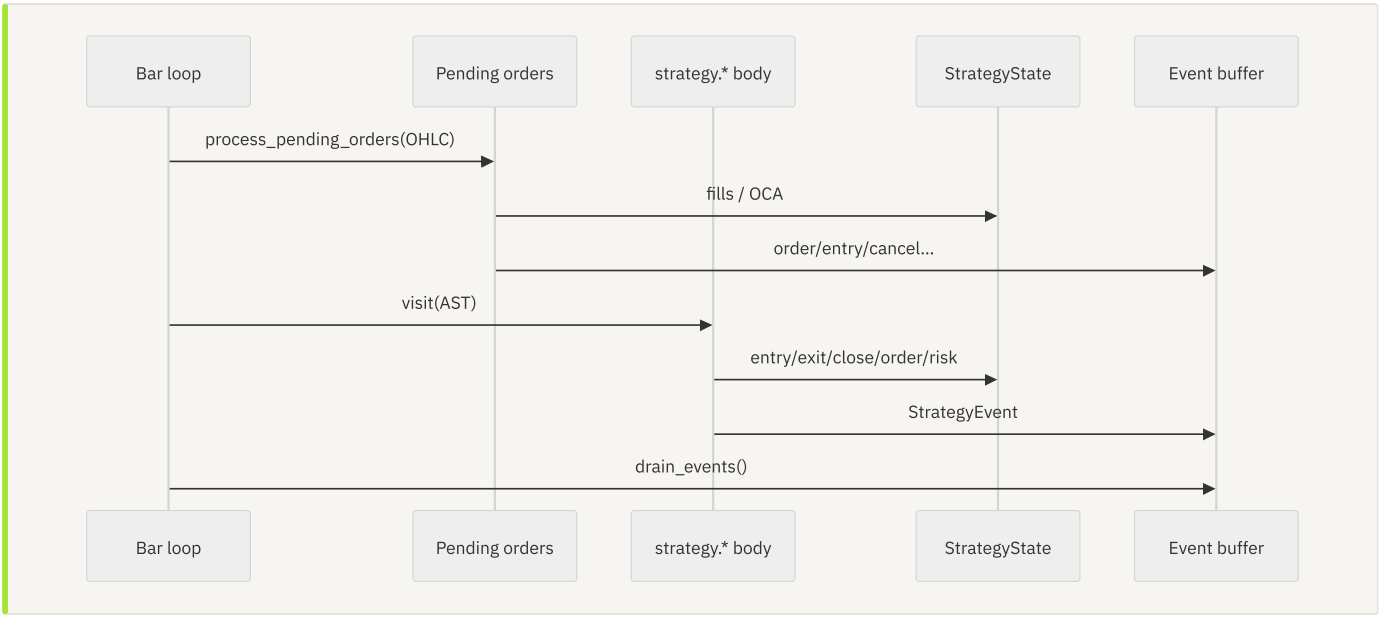
title: "Strategy builtins" description: "Orders, fills, OCA, commission, risk gates, position metrics, and StrategyState."

STRATEGY BUILTINS

Abstract

PYNE's strategy layer is a **per-run broker simulation** driven by `strategy.*` calls during the bar loop. It is not a live exchange adapter: orders become fills against bar OHLC (market immediately; limit/stop via pending book), commissions and slippage adjust PnL, and risk helpers can block entries. Structured StrategyEvent records form the parity contract with pyne-worker and the HOOX trade mesh.

Conceptual model



Fill-before-script matches the interpreter `Runtime` and the compile-path strategy broker.

Interface surface

Order placement

Builtin	Role
`strategy.entry`	Open/add (or reverse) by direction; market or pending limit/stop
`strategy.exit`	Bracket-style exit (pending limit/stop + OCA); market exit when no levels
`strategy.close` / `strategy.close_all`	Flatten by id or all (`qty_percent` supported)
`strategy.order`	Lower-level order with OCA name/type, partial fill cap
`strategy.cancel` / `strategy.cancel_all`	Remove pending orders

Directions accept Pine constants (`strategy.long` / `strategy.short`) and common string aliases.

`strategy.exit` surface (0.3.4+)

Feature	Status
Pending stop/limit brackets + OCA	Interpret + compile — fills via `process_pending_orders` OHLC path
`from_entry` multi-leg filter	Interpret + compile — unknown id is soft no-op
`qty_percent`	Interpret + compile `close` — `%` of target (whole pos or `from_entry`); wins over `qty` ; `≤0` /na → no-op; `>100` capped
Trail (`trail_offset` / `trail_points` / optional `trail_price`)	Interpret + compile — distances are **ticks** (`× mintick`). Pending stop ratchets from bar high/low (OHLC approx). `trail_points=0` / `na` is ignored so a valid `trail_offset` still applies (0.3.8).
`profit` / `loss`	**Ticks** from the exit's entry average (`ticks * mintick`). Long: `entry ± offset` ; short flips the sign. `na` / `≤0` ignore that leg. Absolute `limit` / `stop` win when both are set.

```
//@version=6
strategy("Exit demo", overlay=true, initial_capital=100000)
if bar_index == 5
    strategy.entry("L", strategy.long, qty=10)
if bar_index == 6
    strategy.exit("X", from_entry="L", stop=low * 0.98, limit=high * 1.02, qty_percent=50)
// Tick form (same surface): profit/loss are offsets from entry avg, not prices.
// strategy.exit("X", from_entry="L", profit=200, loss=100)
```

Position and performance series

Zero-arg builtins include `strategy.position_size` (signed: +long / −short), `position_avg_price`, `position_entry_name`, `opentrades` / `closedtrades` counts, `netprofit`, `openprofit`, `equity`, `cash`, gross win/loss stats, averages, max drawdown/runup, and max contracts held.

`strategy.position_avg_price` is the VWAP of **open** entry fills. A filled position is **not** overwritten when `strategy.entry` is called again with the same id (`if cond: strategy.entry("L", strategy.long)` stays at the first fill when `pyramiding=0`). Extra same-direction fills only occur when `opentrades < pyramiding + 1` (PYNE cap = `pyramiding + 1`; TV `pyramiding` default 1 ≡ PYNE 0). Pending limit/stop orders with that id are still replaced. When flat the series is `na`.

Trade queries

Indexed accessors:

- `strategy.closedtrades.entry_*` / `exit_*` / `profit` / `size` / `commission`
- `strategy.opentrades.entry_*` / `size` / `profit` / `commission`
- Open/closed trade fields include bar/time/id/comment plus approximate MAE/MFE (`max_drawdown` / `max_runup` from bar high/low)

Risk (interpret + compile halt cascade)

Builtin	Effect
`strategy.risk.max_position_size`	Cap size as % of equity
`strategy.risk.max_intraday_loss`	Intraday loss gate — percent-of-initial-capital default, `strategy.cash` absolute (0.6.0, both brokers)
`strategy.risk.max_intraday_filled_orders`	Order count cap (day-scoped, enforced on both brokers)
`strategy.risk.max_drawdown`	Absolute / percent drawdown halt
`strategy.risk.max_cons_loss_days`	Consecutive losing calendar days → `entries_blocked`
`strategy.risk.allow_entry_in`	`"all"` | `"long"` | `"short"`

Blocked entries still emit a diagnostic-style event with comment `risk_blocked` where implemented. Compile broker shares the common halt cascade above (not a full TradingView® risk engine).

Declaration

`strategy(title, …)` applies broker settings onto `StrategyState`: `initial_capital`, commission type/value, slippage ticks, pyramiding, `avg_price_model`, etc.

Average price model (pynescript extension)

Not part of official TradingView Pine. Controls how `strategy.position_avg_price` evolves when size changes:

`avg_price_model`	On same-direction adds	On partial reduce
`"stock"` (default)	Arithmetic VWAP of fills	Re-average remaining open legs (FIFO multi-trade style)
`"futures"`	Same arithmetic VWAP	**Sticky** net AEP until flat (linear USDT-M / BTCUSDT-like)
`"inverse"`	Arithmetic today; harmonic add planned	Sticky until flat (same reduce rule as futures)

```
strategy("Perp style", avg_price_model="futures")
// or tokens:
// strategy(..., avg_price_model=strategy.avg_price_futures)
```

Linear add formula (stock and futures):

[avg' = \frac{avg \cdot |size| + p \cdot q}{|size| + q}]

Under `"futures"`, partial closes realize PnL against the sticky average (not per-leg FIFO entry prices), and closed-trade `entry_price` records that same average. Commission is never folded into the average numerator.

There is no `strategy.order.avg_future_price` series—use `strategy.position_avg_price` with the mode switch.

Leverage (pynescript extension — simpler futures UI)

Exchanges expose **leverage** (e.g. 5× / 10× / 20×). Prefer that over TV-style `margin_long` / `margin_short` percentages:

```
//@version=6
// TV-safe: strategy() kwargs need *const* (literals / const vars).
// Host UI can still inject runtime input overrides into PYNE.
strategy(
     "BTCUSDT style",
     avg_price_model="futures",
     leverage=10,
     default_qty_type=strategy.percent_of_equity,
     default_qty_value=100,  // use full equity as *margin*
     initial_capital=10_000)

// Optional Inputs-tab control for *trade logic* (not TV strategy Properties):
// lev = input.float(10, "Leverage", minval=1, maxval=125)
```

Setting	Effect
`leverage=N` (default `1`)	Buying power multiplier
`percent_of_equity` / `cash` default qty	`qty = margin × leverage / price`
`fixed` default qty	Unchanged (contracts as written)
`strategy.cash` / capital held	Margin locked = notional / leverage
`strategy.margin_liquidation_price`	Simple isolated estimate when `leverage > 1`
`strategy.leverage`	Read-back of configured multiplier

`leverage=10` also sets internal margin % to `100/10 = 10` (TV `margin_long` / `margin_short` equivalent). If only `margin_long` is set, leverage is derived as `100 / margin_long`.

`input` before `strategy()` — TV vs PYNE

	TradingView	PYNE
Statements before `strategy()`	Allowed for **const** vars (and types/enums/imports)	Allowed
`input.*()` return qualifier	**`input`** (not `const`)	Plain Python value (no qualifier system)
`strategy(leverage=input.float(...))`	**Compile error** — most `strategy()` params require **`const`**	**Interpret:** works. **Compile:** folds **const-like defval** into the broker ctor (`input.float(10)` → `leverage=10`)
User-adjustable leverage on TV	**Settings** → **Properties** ("Long/Short leverage", mapped from `margin_long` / `margin_short`)	Declaration `leverage=` or host-injected inputs

Qualifier hierarchy on TV: `const < input < simple < series`. A parameter that needs `const` **cannot** accept an `input` value. So even if `lev = input.float(10)` appears *before* `strategy()`, you still cannot pass `lev` into `strategy(...)` on TV.

PYNE does **not** enforce qualifiers; for compile-path parity with TV's const rule, non-const strategy kwargs are skipped unless their series folds to a constant defval.

Internals

`StrategyState` (`strategy.py`)

Per-evaluator instance fields include:

- Position: direction (`flat` / `long` / `short`), size (non-negative internal), entry metadata
- Books: `pending_orders`, `open_trades`, `closed_trades`
- Economics: capital, commission model, slippage, mintick
- Risk: max position %, drawdown, consecutive loss days, `entries_blocked`
- Equity curve peak/trough for max DD / runup
- `_events: list[StrategyEvent]` drained each bar

`signed_position_size()` implements Pine's signed `strategy.position_size`.

`Order` and fills

```
Order: market | limit | stop | stop-limit
  + oca_name / oca_type ∈ {none, cancel, reduce}
  + max_fill_per_bar (0 = fill remaining)
```

`process_pending_orders(open, high, low, close)` walks the book, computes trigger prices from OHLC (gap-aware open logic for limits/stops), applies commission/slippage, updates trades, runs OCA side effects, and emits events.

OCA

oca_type	On fill of a group member
cancel	Cancel other pending in group
reduce	Reduce sibling quantities by fill size
none	Independent

Covered by `tests/test_oca_commission.py` and order-fill suites.

Invariants & edge cases

1. **Isolation.** Never use class-level strategy state; concurrent runs must not share books.
2. **Pyramiding.** Additional entries respect `pyramiding` from the declaration.
3. **Partial fills.** `max_fill_per_bar` / defaults allow multi-bar completion of large orders.
4. **Calendar buckets for cons-loss.** Exit timestamps in ms vs seconds vs bar index are normalized in `note_closed_trade_day`.
5. **Compile path.** Object-mode uses `CompileStrategyBroker`—aligned fill rules, smaller surface than full interpreter metrics. Prefer interpret mode for full `strategy.closedtrades.*` analytics unless compile coverage is verified.

Worked examples

Market long / close

```
//@version=6
strategy("Demo", overlay=true, initial_capital=100000)
if bar_index == 5
    strategy.entry("L", strategy.long, qty=10)
if bar_index == 10
    strategy.close("L", qty_percent=100)
```

Parity fixtures under `tests/fixtures/parity/pine/` encode expected event sequences for such scripts.

Pending stop entry

```
//@version=6
strategy("Breakout", overlay=true)
strategy.entry("Break", strategy.long, stop=high[1])
```

Creates a pending stop; subsequent bars' `process_pending_orders` fill when `high` trades through the stop, using open-gap-aware fill prices.

OCA bracket sketch

```
//@version=6
strategy("Bracket", overlay=true)
// after entry "L" is open:
strategy.order("TP", strategy.short, qty=1, limit=tp, oca_name="br", oca_type=strategy.oca.cancel)
strategy.order("SL", strategy.short, qty=1, stop=sl, oca_name="br", oca_type=strategy.oca.cancel)
```

First fill cancels the sibling.

Failure modes

Symptom	Cause
Entry never fills	Pending stop/limit; OHLC never trades through; or risk block
Double entries	Pyramiding > 0 or missing close
Events missing `script_id`	Host forgot to stamp after `drain_events` (Runtime does this)
PnL off vs TV	Commission type, slippage ticks, mintick, or fill price model
Interpret vs compile event drift	Broker subset / timing—diff with `tests/test_compiler_strategy.py`

See also

- Events
- Strategy broker (compile)
- pyne-worker
- Pro API backtest

COLLECTIONS

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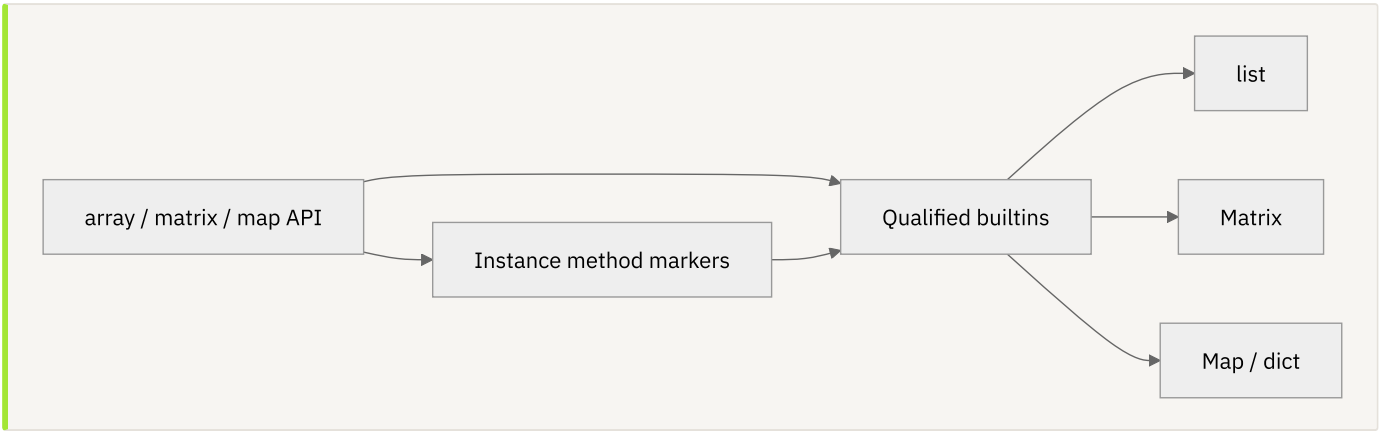
title: "Collections" description: "array.*, matrix.*, and map.* runtime semantics, method dispatch, and type tags."

COLLECTIONS

Abstract

Pine collections—**arrays**, **matrices**, and **maps**—are mutable values that live in the evaluator context (often under `var` so identity persists across bars). PYNE maps them to Python `list`, a dedicated `Matrix` type, and a `Map` / `dict`-like type respectively, with builtin functions and instance-method sugar (`a.push(x)` → `array.push(a, x)`).

Conceptual model



Interface surface

Arrays (`array.*`)

Represented as Python lists. Highlights from the dispatch map:

Category	Functions
Construct	`array.new` , `array.new_<type>` , `array.from`
Mutate	`push` , `pop` , `shift` , `unshift` , `insert` , `remove` , `set` , `fill` , `clear` , `reverse` , `sort`
Access	`get` , `first` , `last` , `size` , `includes` , `indexof` , `lastindexof` , `slice`
Aggregate	`sum` , `avg` , `min` , `max` , `median` , `mode` , `range` , `stdev` , `variance` , covariance, percentiles, percentrank
Search	`binary_search` , `binary_search_leftmost` , `binary_search_rightmost` — UDT arrays take `sort_field` (const int index, default 0, or const string name; array must be sorted by that field ascending)
Higher-order	`every` , `some` , `copy` , `concat` , `join` , `abs`

Indexing: Pine array indices are **0-based from the start of the list** (unlike series history). Negative indices are rejected for series; array paths use list semantics consistent with the builtin handlers.

UDT `sort_field` (0.3.6 / Pine August 2026): `array.sort` , `array.sort_indices` , and `array.binary_search*` compare one field of user-defined type elements. Pass a const int field index (`0` is the first declared field, and the default for search) or a const string field name. Binary search requires the array to already be sorted by that field in ascending order. Method form (`a.binary_search(value, "id")`) and `sort_field=` kwargs work on interpret and compile.

Series-like values passed into array ops unwrap via `.history` (reversed to chronological) when duck-typed.

Matrices (`matrix.*`)

`Matrix` supports:

- Construction: `matrix.new`
- Element access: `matrix.get` / `set` , and `m[row, col]` via subscript
- Shape: `rows` , `columns` , `elements_count`
- Row/column ops: add/remove/copy, sum/avg/min/max/mode, fill

- Aggregates: sum/avg/min/max/mode all
- Transforms: `transpose`, fill diagonal, and further linear helpers in the evaluator map

Method sugar: `m.rows()` → `matrix.rows(m)`.

Maps (`map.*`)

Function	Role
`map.new`	Empty map
`put` / `put_all` / `get` / `remove` / `clear`	Mutation
`contains` / `keys` / `values` / `size` / `copy`	Query

Keys follow Pine map rules as implemented (hashable Python keys). Compile object-mode lowers maps to ordinary Python `dict`s.

Internals

Path	Role
`src/pynescript/ast/evaluator/builtins/arrays.py`	Array builtins
`src/pynescript/ast/evaluator/builtins/matrix.py`	`Matrix` type
`src/pynescript/ast/evaluator/builtins/matrix_evaluator.py`	matrix.* handlers
`src/pynescript/ast/evaluator/builtins/map.py`	`Map` type
`src/pynescript/ast/evaluator/builtins/map_evaluator.py`	map.* handlers
`src/pynescript/ast/evaluator/names.py`	Method markers for list/Matrix
`tests/test_collections*.py`, `test_map_collections.py`, `test_matrix_*.py`	Coverage

Multi-dispatch type tags (`array.float`, `matrix.string`, …) interact with method overloading and `na` exclusion lists so `na`-typed optionals do not bind to collection `tostring` overloads.

Invariants & edge cases

1. **Identity under `var`.** `var a = array.new_float(0)` keeps one list across bars; reassigning `a := array.new_float(0)` replaces it.
2. **Drawing-typed arrays.** `array.new_label` etc. store drawing object references; delete semantics remain drawing-registry concerns.
3. **Empty array overload match.** Empty arrays match any `array.*` element tag in dispatch (no sample element).
4. **Matrix unpack hazard.** StatementEvaluator refuses to treat matrices as general iterables for tuple unpack—prevents corrupting multi-assign from row iteration.
5. **Compile mode.** Maps/UDTs force **object mode**; pure array numeric work may still numeric-compile depending on visitor detection—verify generated code when performance-critical.

Worked examples

Rolling buffer

```
//@version=6
indicator("buf")
var floats = array.new_float(0)
array.push(floats, close)
if array.size(floats) > 20
    array.shift(floats)
plot(array.avg(floats))
```

Matrix element

```
var m = matrix.new<float>(2, 2, 0.0)
matrix.set(m, 0, 1, close)
plot(matrix.get(m, 0, 1))
```

Map counter

```
var counts = map.new<string, int>()
map.put(counts, "n", nz(map.get(counts, "n")) + 1)
```

Binary search on a UDT array

```
//@version=6
indicator("udt search")
type Item
    int id
    float v
var a = array.new<Item>()
if bar_index == 0
    array.push(a, Item.new(2, 20.0))
    array.push(a, Item.new(1, 10.0))
    array.push(a, Item.new(3, 30.0))
    array.sort(a, order.ascending, "id")
plot(array.binary_search(a, 2, "id"))
plot(a.binary_search_leftmost(2, sort_field="id"))
```

Failure modes

Symptom	Cause
`map.get requires map and key`	Wrong arity or non-Map receiver
Index errors	OOB get/set without Pine `na` guard in that handler
Method not found	Typo or non-list receiver after series unwrap failed
Shared mutation across runs	Reused evaluator context without reset

See also

- Expressions & statements
- Drawing & plotting — arrays of labels/lines
- Compiler object mode

DRAWING PLOTTING

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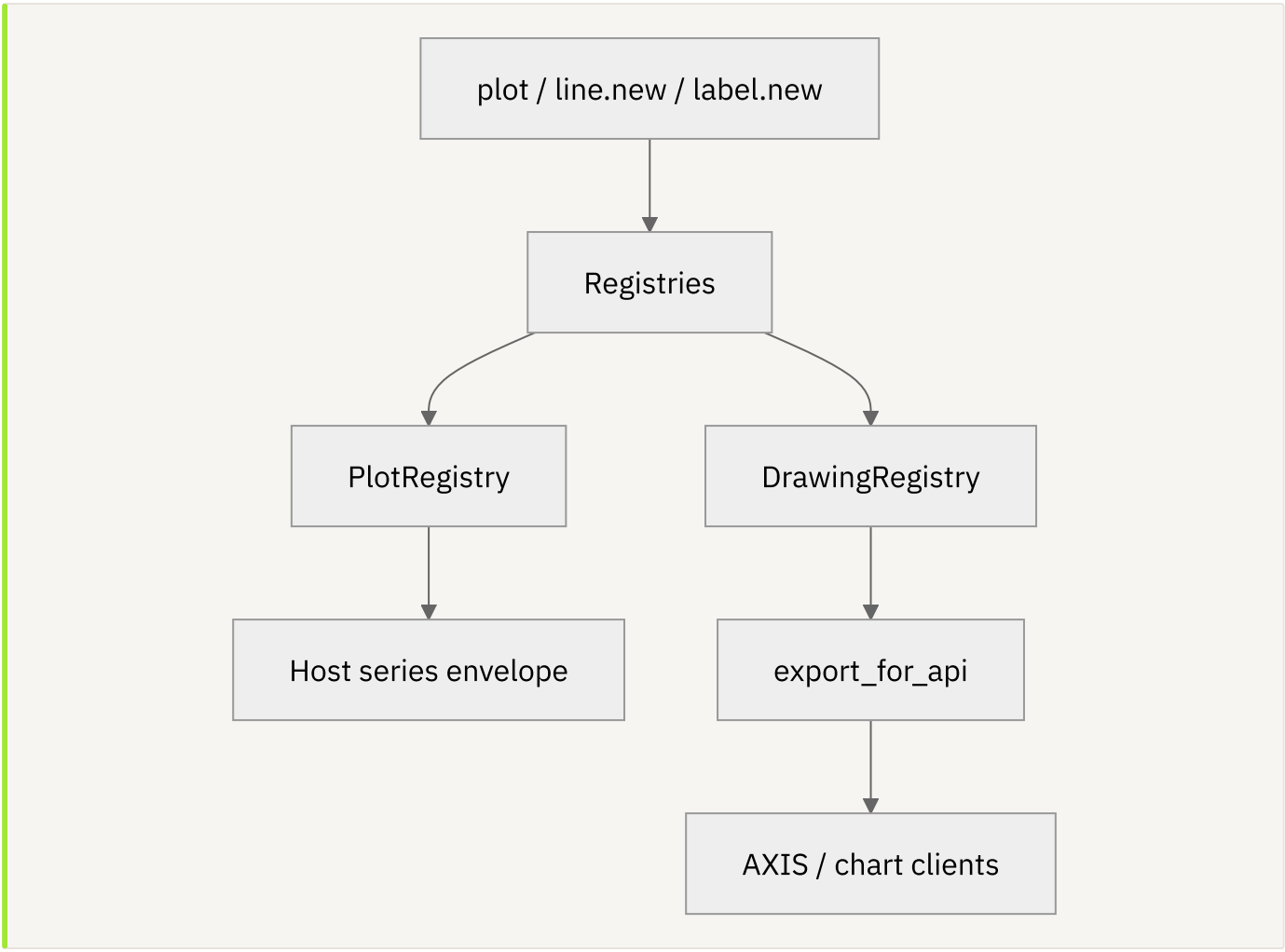
title: "Drawing and plotting" description: "plot* side effects, DrawingRegistry objects, and export shapes for hosts and AXIS."

DRAWING AND PLOTTING

Abstract

Visual effects in Pine are **side channels**: they do not change pure arithmetic results, but they are first-class runtime outputs. PYNE records plots in `PlotRegistry` and drawing primitives (lines, boxes, labels, tables, polylines, linefills) in `DrawingRegistry`, then serializes them for the Pro API and optional AXIS. Evaluation never requires a UI—tests assert on registries alone.

Conceptual model



Interface surface

Plotting (`PlottingFunctionsMixin`)

Function	Kind	Notes
`plot`	`plot`	Returns `Plot` id for `fill(plot1, plot2)` ; series values in host export
`plotshape` / `plotchar` / `plotarrow`	markers	Condition series + `plot_meta.kind` (style/location/char). Visuals are **not** in `drawings`
`barcolor` / `plotbar` / `plotcandle`	overlays	Per-bar color / OHLC series + kind stamp (0.3.12 dual-host)
`hline`	horizontal	**Both** hosts export a **constant price series** keyed by title + `plot_meta.kind: "hline"` . Not in `drawings`
`bgcolor` / `barcolor`	coloring	Color-string series + `plot_meta.kind` . Not in `drawings`
`fill`	fill	**Both** export a **titled series key** (values often all-null); band color + plot refs live in `plot_meta`

Style constants: `plot.linestyle_solid` / `dashed` / `dotted` . Plot style tokens: `plot.style_line` , `linebr` , `stepline` , `steplinebr` , `stepline_diamond` , `histogram` , `columns` , `cross` , `area` , `areabr` , `circles` .

Each call constructs a `Plot` dataclass (`kind`, `series`, `title`, `color`, linewidth, text formatting, force_overlay, ...) and appends it to `PlotRegistry.plots`.

Hosts often **also** capture per-bar plot values on the evaluator (`plot_outputs` / `_plot_value_cols`) for time-aligned series maps—see `src/pynescript/runtime/evaluator.py` and packaging in `src/pynescript/runtime/host.py` (`backend.evaluator` / `backend.runtime` re-export those modules).

`PYNE_LIGHT_PLOTS=1` skips columnar capture, plot-registry handles, and `input.*` metadata (corpus OK/fail only).

Title defaults and series keys

Runtime packages plots into `result["series"]` / `result["plot_meta"]` on **both** hosts. `result["drawings"]` is **geometry-only** (line / box / label / polyline / table / linefill). Compile `__drawings` is an internal event list that Runtime **folds and strips** before JSON — hosts never need it. Empty or missing titles get defaults; collisions get `_2`, `_3`, ... suffixes so map keys stay unique.

Call	Default title	Notes
`plot`	`plot_0`, `plot_1`, ...	Compile emits `plot_{n}` at emit time; interpret empty title → `plot_{index}` at JSON packaging
`hline`	`hline`	Uniquified `hline`, `hline_2`, ... when untitled levels repeat
`fill`	`fill`	Same uniquify rule (`Background` / `Background_2`, ...)
`bgcolor`	`bgcolor`	Series + `plot_meta` (both hosts)
`plotshape`	`shape`	Series + `plot_meta` (both hosts)

Explicit `title=` (or positional title) wins when present.

`hline` — constant series (both modes)

Interpret (`src/pynescript/runtime/evaluator.py` + Runtime packaging):

- Captures kind `hline` with the price as the series cell each bar.
- JSON export **forward-fills** nulls with the known constant so AXIS can draw a full-width level.
- `plot_meta[title]` includes `kind: "hline"`, `price`, `color`, `linewidth`, ...

Compile:

- Emits a **plot array** (`safe_float(price)` every bar) under the uniquified title and stamps `plot_meta.kind: "hline"`.
- May also append an internal `__drawings` event; Runtime packaging **strips** visual kinds so host `drawings` stays geometry-only.
- Host wraps NaN → null; constant levels remain numeric across the bar range.

Parity tests: `tests/test_multi_plot_cross.py` (`test_hline_series_keys_compile_matches_interpret`), `tests/test_plot_drawing_dual_host.py`, `tests/test_compiler_objects.py`.

`fill` — titled series key for key-set parity (both modes)

Interpret:

- Meta stores sibling series titles as `plot1` / `plot2` (resolved from `plot()` handles).
- Capture may hold a color string on the column; **Runtime JSON packaging** typically emits **all-null** series cells for fills (non-numeric / non-bgcolor strings → `null`). Band **color** and plot refs stay in `plot_meta` (and registry fill

objects when ids are enabled).

Compile:

- Registers a series key with expression `None` (float64 NaN → JSON `null`) under the uniquified title so **series key sets match interpret**.
- Band details (plot refs, color) land in `plot_meta`. An internal `__drawings` `fill` event, if emitted, is **stripped** before Runtime JSON.

Parity tests: `test_fill_background_series_keys_compile_matches_interpret`, `test_fill_title_series_key_and_drawings`, `tests/test_plot_drawing_dual_host.py`.

Import stubs must not leak as series strings

Unresolved `import …` libraries become chainable stubs marked `__pine_import_stub__ = True` (repr like `<PineImportStub namespace/name/version>`). If a script plots a stub member, Runtime's `_json_plot_value` maps those cells to **`null`**—never the stub repr or a color-like string—so AXIS and interpret/compile comparisons never see `"<PineImportStub …>"` as a plot series value.

Drawing objects (`DrawingBuiltinsMixin`)

Object types: `Line`, `Box`, `Label`, `Table`, `Polyline`, `LineFill`, `ChartPoint`.

Typical factories: `line.new`, `box.new`, `label.new`, `table.new`, `polyline.new`, `linefill.new`, plus getters/setters/deletes and `*.all` / last-bar helpers where implemented.

Instance methods resolve through namespace markers:

```
la.get_text()  →  label.get_text(la)
```

`DrawingRegistry.export_for_api(bar_times)` maps `xloc=bar_index` coordinates to wall times for chart clients, normalizes colors, and skips deleted objects. `xloc.bar_index` **past the last bar** (classic `bar_index + 1` on `barstate.islast`) is extrapolated from the series period (0.3.5); empty `bar_times` passes the bare bar index.

Line / box / label payloads include **`force_overlay`** for AXIS pane routing (0.3.6). `linefill.new` serializes as `type: "linefill"` quads (`t1` / `p1` … `t4` / `p4` from the two line endpoints, plus `color` / `bgcolor`).

Registry lifecycle

```
DrawingRegistry.reset()  # clears drawings + PlotRegistry
```

Hosts must reset at run start so labels from a previous evaluation do not leak.

Drawing GC (`max_*_count`)

Interpret path enforces TradingView®-style **garbage collection** on drawing factories (`line.new`, `label.new`, `box.new`, `polyline.new`, …):

Declaration kwargs (`indicator` / `strategy`)	Default	Hard cap
`max_lines_count`	50	500
`max_labels_count`	50	500
`max_boxes_count`	50	500
`max_polylines_count`	50	100

When more **active** objects of a type exist than the cap, the **oldest** are marked `deleted=True` so they leave `*.all` and `DrawingRegistry.export_for_api`. Caps are applied from the script declaration at run start (`DrawingRegistry.configure_from_declaration`); shrinking a cap immediately collects.

```
//@version=6
indicator("labels", overlay=true, max_labels_count=100)
if barstate.islast
    label.new(bar_index, high, "x")
```

Compile object-mode drawings are a simplified event list—GC caps apply on the **interpret** registry path; prefer interpret when you need full last-bar / `*.all` / cap behavior.

`fill` + `plot_meta` for AXIS

For dual `plot` + `fill` bands:

1. Each `plot()` contributes a **numeric series** under its title (default `plot_0`, ...).
2. `fill(p1, p2, …)` contributes a **titled series key** (often all-null cells) so key-sets stay stable, plus `plot_meta` that carries band color and plot refs.
3. Hosts such as AXIS should read `plot_meta[title]` for `kind`, colors, and sibling plot titles (`plot1` / `plot2`) rather than treating fill series as prices.

Interpret packaging lives in `src/pynescript/runtime/host.py` / `evaluator.py`; compile emits matching series keys. Internal `__drawings` fill events are folded/stripped before Runtime JSON.

Internals

Path	Role
`src/pynescript/ast/evaluator/builtins/plotting.py`	`Plot`, `PlotRegistry`, plot* handlers
`src/pynescript/ast/evaluator/builtins/drawing.py`	Drawing types, registry, builtins, `export_for_api`, `fold_compile_drawing_mutations`
`src/pynescript/runtime/evaluator.py`	Host plot capture: hline / fill / bgcolor / plotshape series + meta
`src/pynescript/runtime/host.py`	JSON packaging: series keys, stub nulling, hline fill-forward, fill plot refs
`src/pynescript/compiler/compiler.py`	`hline` / `fill` series + internal `__drawings`; visual events lifted then stripped
`src/pynescript/ast/evaluator/names.py`	`_DRAWING_METHOD_NS`
`tests/test_plotting_effects.py`, `test_drawing_all_and_last_bar.py`, `test_multi_plot_cross.py`, `test_bgcolor_plotshape_export.py`, `test_plot_drawing_dual_host.py`, `test_compiler_objects.py`	Behavior / parity

Compile object-mode notes

Geometry factories and some marker-style calls force **object mode** (pure-Python bar loop with an append-only `__drawings` event list). Runtime packaging **lifts** visual events into `series` + `plot_meta.kind`, then **strips** them so host `drawings` is geometry-only — the same AXIS shape as interpret `export_for_api`.

Call	Host export (both modes)	Compile internals
`hline`	Constant series + `plot_meta.kind`	Series array; optional `__drawings` event stripped
`fill`	All-null series key + `plot_meta` color/refs	Series key; optional `__drawings` event stripped
`plotshape` / `plotchar` / `plotarrow`	Condition/signed series + `plot_meta.kind`	Visual events lifted into series, then stripped
`bgcolor` / `barcolor`	Color-string series + `plot_meta.kind`	Same lift/strip as other visuals
`plotbar` / `plotcandle`	OHLC series + kind stamp	Same lift/strip
`label` / `line` / `box` / `polyline` / `table` / `linefill`	Geometry objects in `drawings`	Handle dict + `__drawings`; `set_*` / `*.delete` fold via `DrawingRegistry.fold_compile_drawing_mutations`

Numeric mode still supports plain `plot` float64 arrays; geometry or marker calls can flip object mode for that script.

Invariants & edge cases

1. **`plot` returns an id.** Required for `fill`; treating it as “void” breaks band fills. Hosts that skip `PlotRegistry` unless `fill(` appears in source must still soft-fail fill gracefully.

2. **Deleted flag.** Soft-delete keeps history for debugging; `active()` filters deleted plots and GC-collected drawings.
3. **`max_*_count` GC.** Oldest active drawings are deleted when caps are exceeded (interpret registry).
4. **Series in drawings.** Prices may be series wrappers—export coerces `.current` and maps NaN → omit.
5. **Compile path.** Numeric mode supports `plot` arrays; geometry/markers force object mode with an internal `__drawings` event list. Runtime **folds** `set_*` / `*.delete`, **lifts** visual events into `series` + `plot_meta.kind`, then **strips** them so `drawings` is geometry-only. Prefer interpret when you need full last-bar / `*.all` / registry GC — not because visuals lack series.
6. **Import stubs → null.** `__pine_import_stub__` and `"<PineImportStub…>"` strings never appear as plot series cells after Runtime packaging.
7. **No GPU/canvas in-process.** Registries are data; AXIS/HOOX render elsewhere.

Worked examples

Dual plot + fill

```
//@version=6
indicator("BB mid")
basis = ta.sma(close, 20)
p1 = plot(basis)
p2 = plot(basis * 1.01)
fill(p1, p2, color=color.new(color.blue, 80))
```

Label on last bar

```
//@version=6
indicator("lbl", overlay=true)
if barstate.islast
    label.new(bar_index, high, str.tostring(close))
```

Hosts read `DrawingRegistry.labels` or API export after the run.

Failure modes

Symptom	Cause
Empty drawings in API	Forgot `DrawingRegistry.reset` timing / export; or script never called `*.new`
fill no-ops	plot ids not retained from `plot()` returns
Stale labels across HTTP runs	Missing registry reset between `Runtime.run` calls (Runtime resets—custom hosts must)
Object mode missing drawing	Compile still folds `set_*` / `delete`; leftover events usually mean fold did not run
Only 50 labels/lines visible	Default `max_*_count`; raise on `indicator` / `strategy` (≤ hard cap)
fill series all null	Expected for band keys; use `plot_meta` for color + plot refs

See also

- Builtins hub

- Pro API preview / chart renderer
- AXIS docs
- Compiler overview

REQUEST INPUT

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title: "request.* and input.*" description: "Multi-symbol/timeframe requests, fundamental na semantics, host series bind, and input parameter resolution."

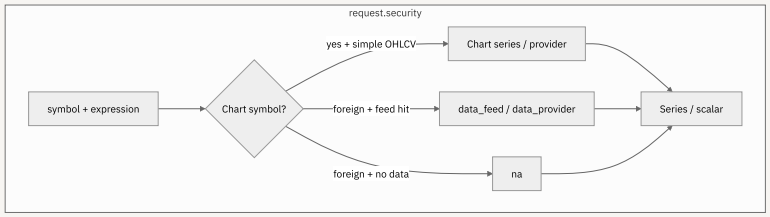
`request.*` AND `input.*`

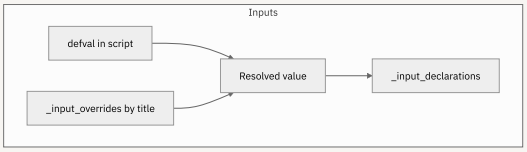
Abstract

Two namespaces couple scripts to the **host environment**: `input.*` declares parameters (with defaults and UI metadata), and `request.*` pulls data from other symbols, timeframes, or fundamental/economic sources. PYNE implements both as builtins with explicit extension points— `data_feed` , `data_provider` , and `_input_overrides` — so the same script can run offline or online with real market data.

Design rule: missing multi-symbol or fundamental data yields **`na`** , never silent substitution of chart OHLCV (no close-as-dividend, no chart volume as UPVOL).

Conceptual model





Interface surface

`input.*` (`InputBuiltinsMixin`)

Builtin	Purpose
`input`	Generic defval + title/tooltip/inline/group/confirm/active
`input.bool` / `int` / `float` / `string` / `color`	Typed scalars
`input.price` / `source` / `time` / `timeframe` / `session` / `symbol`	Domain inputs
`input.enum` / `input.text_area`	Enumerations and multiline text

Runtime value: each call returns the resolved value (override if title matches, else default). Metadata is appended to `_input_declarations` for settings panels and LSP-adjacent hosts.

Overrides live on the evaluator as `_input_overrides: dict[title, value]`.

`request.*` (`RequestBuiltinsMixin`)

Builtin	Role
`request.security`	Other symbol / timeframe expression
`request.security_lower_tf`	Lower-TF array expansion
`request.dividends` / `earnings` / `splits`	Corporate actions
`request.financial` / `economic` / `quandl`	Fundamentals / external series
`request.currency_rate`	FX conversion helper
`request.seed`	Deterministic pseudo-series
`request.footprint`	Volume footprint object (v6 surface)

`request.security` — interpret path

Resolution order (simplified):

1. Normalize symbol (series → last element; `ticker.*` → symbol string) and timeframe.
2. Classify **chart vs foreign** (`_is_chart_symbol` vs host `syminfo` / provider symbol).
3. Try `data_feed.fetch_latest_ohlcv` / ticker, then `data_provider.fetch`.
4. **Same-symbol coarser timeframe (HTF resample, 0.3.4+):** bucket chart OHLCV to the requested TF; **last completed** HTF bar by default (lookahead_off-style). Allowlisted simple TA on HTF series — `ta.sma` / `ta.ema` /

`ta.rsi` / `ta.atr` / `ta.wma` / `ta.rma` (bare `close` / `high` /... source + const length; `ta.atr(n)` length-only) — evaluates on the unique completed HTF series. Nested `ta.*`, UDFs, and non-const lengths stay `na`. `barmerge.gaps_on` / `lookahead_on` are honored on these resample paths (0.6.0, finalized-bucket lookahead in 0.6.1): gaps deliver the value only on bucket-start bars, lookahead leaks each HTF bucket's final value from the period start on historical bars. Both stay unused on passthrough / provider / complex-na paths. Policy meta is exposed on `result["meta"]["request_security"]` (`gaps_supported` / `lookahead_supported`, `gaps_applied` / `lookahead_applied`).

5. **Foreign + pre-evaluated expression** (UDF result, list/tuple of chart values, non-string expr) without multi-symbol data → **`na`** (not chart close as “dividends”).
6. Fundamental / non-equity prefixes (`DIVIDEND`, `FACTSET`, `EARNINGS`, `ESD_`) with no feed hit → **`na`** (no mock OHLCV).
7. Bare equity-style string names may still use **legacy mock** prices for offline demos when no feed is wired.

`ChartOHLCVProvider` (wired by `Runtime` via `resolve_request_sources` when unset) **only serves the chart ticker**. Foreign tickers get empty series, so interpret returns `na` unless a real multi-symbol feed is configured.

`request.security` — compile path

Compile lowering is intentionally narrow (compiler overview):

Case	Emit
Same-symbol (`syminfo.ticker` / empty chart id) **and** simple OHLCV expr (`close`, `high[1]`, ...)	Passthrough chart array sample
Foreign tickers (`UPVOL.NY`, `ESD_FACTSET`, ...)	`np.nan`
Complex third arg (UDFs, `year_sum(close)`, non-allowlisted `ta.*`, arithmetic)	`np.nan`
Other `request.*` APIs	`np.nan` (object mode)

No inventing chart-close-as-dividends or chart-volume-as-advance/decline series.

Footprint (`FootprintBuiltinsMixin`)

Types `Footprint` and `VolumeRow` expose buy/sell volume, delta, VAH/VAL/POC rows, and per-row imbalance helpers—aligned with the v6 footprint surface inventory.

Mode selection (`mode=auto`)

Pro API `/run` defaults body `mode` to **`auto`**. `Runtime._compile_eligible` **rejects compile** when the source contains `request.` (or top-level `import`), so auto **prefers interpret** for any script that uses `request.*`:

```
request.* present → compile_fallback_reason = "request.* not supported in compile path"
                  → auto_backend = interpret
```

Forced `mode=compile` still runs the narrow lowering above (simple same-symbol only; else `na`). Input overrides also force interpret under auto.

See Compiler overview for warm compile, numeric vs object mode, and eligibility.

Host series bind (interpret)

Assigning host OHLCV / time series into a **user name** never aliases the host buffer by reference. `_bind_series_name` copies the current scalar into a fresh series for the user name so patterns like:

```
last_t := na(close) ? last_t[1] : time
```

cannot corrupt `time[j]` history. That bug previously broke TTM / `year_sum`-style windows (e.g. dividend-yield scripts).

Internals

Path	Role
`src/pynescript/ast/evaluator/builtins/input.py`	Input handlers + declarations
`src/pynescript/ast/evaluator/builtins/request.py`	`request.*` + chart/foreign / na policy
`src/pynescript/ast/evaluator/statements.py`	`_bind_series_name` (no OHLCV alias)
`src/pynescript/util/data.py`	`ChartOHLCVProvider`, `resolve_request_sources`
`src/pynescript/compiler/compiler.py`	Same-symbol OHLCV-only `security` lower
`src/pynescript/runtime/host.py`	Package Runtime SoT; feed wiring; `_compile_eligible` / `_run_auto`; copies policy onto `meta.request_security`
`backend/runtime.py`	Compat re-export of package Runtime (not the implementation)
`tests/test_dividend_yield_parity.py`, `test_datafeed_wiring.py`	na parity + chart provider

Invariants & edge cases

1. **Inputs are pure values at runtime.** Titles matter only for override keys and UI metadata—not for Pine type identity.
2. **Foreign without data → `na`.** Prefer honest missing data over inventing chart series as multi-asset results.
3. **Chart provider is chart-only.** Multi-asset accuracy needs a real `data_feed` / `data_provider`.
4. **Mocks are limited.** Equity-style bare symbols may still mock offline; fundamental prefixes and foreign pre-evaluated exprs do not.
5. **Dynamic symbols.** List/series symbols resolve to the latest element—supports loops constructing ticker ids.
6. **Lower TF.** `request.security_lower_tf` returns array-like structures; length scales with simulated lower-TF density when mocking.
7. **`mode=auto` + `request.*` → interpret.** Compile path is not a full multi-asset substitute (`_compile_eligible` rejects `request.`).
8. **HTF is last-completed only.** Forming HTF buckets are never returned; `gaps` / `lookahead` kwargs do not change that.

Worked examples

Parameterized length

```
//@version=6
indicator("len")
len = input.int(14, "Length", minval=1)
plot(ta.sma(close, len))
```

Host:

```
ev._input_overrides = {"Length": 21}
```

Multi-timeframe close / simple HTF TA (chart symbol)

```
//@version=6
indicator("HTF")
htf = request.security(syminfo.tickerid, "D", close)
htf_sma = request.security(syminfo.tickerid, "D", ta.sma(close, 20))
plot(htf)
plot(htf_sma)
```

Same-symbol coarser TF: interpret buckets chart OHLCV (last **completed** HTF bar) and evaluates allowlisted simple `ta.*` (`sma` / `ema` / `rsi` / `atr`) on that series. Compile may passthrough simple same-symbol OHLCV; complex foreign/security still → `na` . Inspect `meta.request_security.policies` (`htf_ohlcv_resample` , `htf_simple_ta_resample` , `foreign_na` , `complex_htf_na` , ...).

Intentional `na` — dividend yield (fundamentals missing)

```
//@version=6
indicator("div")
year_sum(src) =>
    ta.cum(src)
div_ticker = ticker.new("ESD_FACTSET", "X;Y;DIVIDENDS")
div_ttm = request.security(div_ticker, "D", year_sum(close), barmerge.gaps_on, lookahead=barmerge.l
plot(div_ttm)
```

Without a fundamentals feed: **interpret and compile both plot `na`** for `div_ttm` —not chart close as fake TTM dividends. Covered by `tests/test_dividend_yield_parity.py` .

Intentional `na` — CVI / UPVOL-style foreign OHLCV

```
//@version=6
indicator("cvi")
up = request.security("UPVOL.NY", "D", close)
plot(up)
```

Foreign ticker + no multi-symbol data → `na` . Compile never rewrites this as chart close; inventing advance/decline volume from the host series is forbidden.

Failure modes

Symptom	Cause
All-`na` multi-asset plots	Foreign symbol / fundamental expr; no feed (expected)
Always ~100 mock prices	Bare equity string + no feed; legacy mock path
Input override ignored	Title string mismatch (including empty title)
Footprint fields zero	`request.footprint` without configured footprint data
`time[1]` / history wrong after assign	Should be fixed: host bind no longer aliases OHLCV
Lookahead surprises	Host data alignment / gaps—not automatic TV replay guarantees
`auto_backend=interpret` with `request.*`	Eligibility prefilter; use interpret or wire multi-symbol feed

See also

- Compiler overview — auto eligibility, same-symbol security lower
- Builtins hub
- Series & history
- Pro API run endpoint
- Runtime hub

LIBRARIES

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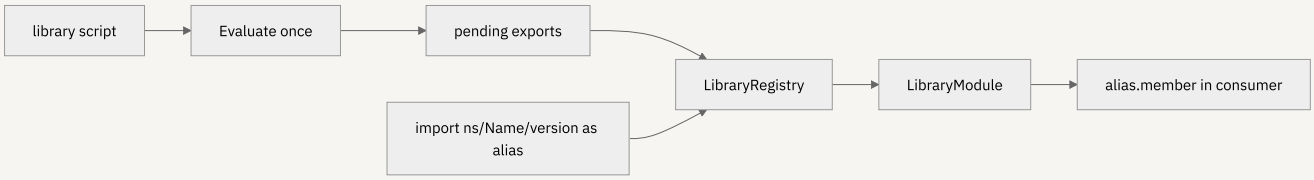
title: "Libraries" description: "In-process library export/import, LibraryRegistry, and evaluate-order requirements."

LIBRARIES

Abstract

TradingView libraries publish as `namespace/Name/version` with `export`ed members. PYNE implements an **in-process** analogue: evaluate a `library("Title")` script (or register source by path), collect exports (`export const`, exported functions, types), then `import` binds an alias to a `LibraryModule` for `alias.member` access. There is no network publish step—the registry is the host's responsibility.

Conceptual model



Interface surface

Library script

```
//@version=6
library("MyMath")

    a + b
```

On script completion, `StatementEvaluator` finalizes `_pending_library_exports` onto the active `LibraryModule` and calls `LibraryRegistry.register`.

Consumer import

```
//@version=6
indicator("use")

plot(M.add(close, M.PHI))
```

Resolution (`LibraryRegistry.lookup`):

1. Exact `(namespace, name, version)` if provided and registered.
2. Else title-only match (`MyMath`) for local evaluation workflows without publisher path.

API on the evaluator

```
ev.register_library_source(namespace, name, version, source_str)
mod = ev.lookup_library(namespace=..., name=..., version=...)
```

`register_library_source` stores Pine text for **lazy** load on the first matching `import`. `_load_library_source` evaluates definitions only (skips showcase body) and **finalizes** `_pending_library_exports` onto the `LibraryModule` so `export const` / exported functions bind (0.3.8). Evaluating a library script on the same evaluator still populates exports eagerly.

Host evaluate (`0.3.7+`)

Hosts pass published sources without a pre-eval step:

```
Runtime(symbol="TEST").run(
    consumer_source,
    ohlcv,
    mode="interpret",  # or "auto" — import is compile-ineligible
    libraries=[
        {"namespace": "ns", "name": "Lib", "version": 1, "source": library_source},
    ],
)
```

The same list is accepted on Pro API and **pyne-worker** `POST /run` as `libraries`. pyne-worker also stores the array on deployed scripts so cron / `script_id` reuse it. Pro API slices to **32** entries (`backend/app.py`); `Runtime.run` itself does not cap. `mode=compile` has no import path — `mode=auto` treats `import` as compile-ineligible and **forwards `libraries=` into interpret fallback** (0.3.9).

`LibraryModule`

Dataclass with `title`, optional `namespace` / `version`, and `exports: dict[str, Any]`. Attribute access reads exports; missing members raise `AttributeError` with a clear message. `NameEvaluator` routes `alias.member` when the base value is a `LibraryModule`.

Internals

Path	Role
`src/pynescript/ast/evaluator/libraries.py`	`LibraryModule`, `LibraryRegistry`
`src/pynescript/ast/evaluator/statements.py`	export registration, import visit, library finalize
`src/pynescript/ast/evaluator/base.py`	registry fields on `BaseEvaluator`
`src/pynescript/ast/evaluator/__init__.py`	`register_library_source`, `lookup_library`
`src/pynescript/runtime/host.py`	`Runtime.run(..., libraries=)` + auto-mode forward
`tests/test_library_export_import.py`	Round-trip coverage

Exportable values include callables (user functions), constants, and type-related bindings depending on declaration paths. Exported methods participate in the same call machinery as local functions once retrieved from the module.

Invariants & edge cases

1. **Evaluate library before consumer** (or register source for lazy import). Empty registry → import failure.
2. **Same evaluator instance** (or shared registry) must see both scripts—registries are not process-global unless the host shares them.
3. **Version is part of the path key.** Mismatched version does not fall back to another version automatically when namespace is specified.
4. **Title registration** always updates `_by_title` so simple local titles work without publisher metadata.
5. **No TV account auth.** Publishing/versioning outside the process is out of scope.

Worked examples

Explicit registration

```
from pynescript.ast.evaluator import NodeLiteralEvaluator

ev = NodeLiteralEvaluator()
ev.evaluate_script(library_source)   # registers by title
ev.evaluate_script(consumer_source)  # import by title or path
```

Path registration without pre-eval

```
ev.register_library_source("user", "MyMath", 1, library_source)
# import user/MyMath/1 loads definitions and finalizes exports on first visit
```

Failure modes

Symptom	Cause
Import resolves to nothing	Library never registered / wrong title
`Library 'X' has no exported member 'y'`	Missing `export` or typo
Stale exports	Re-evaluated library without re-import; host cache
Version conflict	Multiple libs same title—last `register` wins for title key

See also

- Expressions & statements
- Builtins declarations
- End-user library API guide

EVENTS

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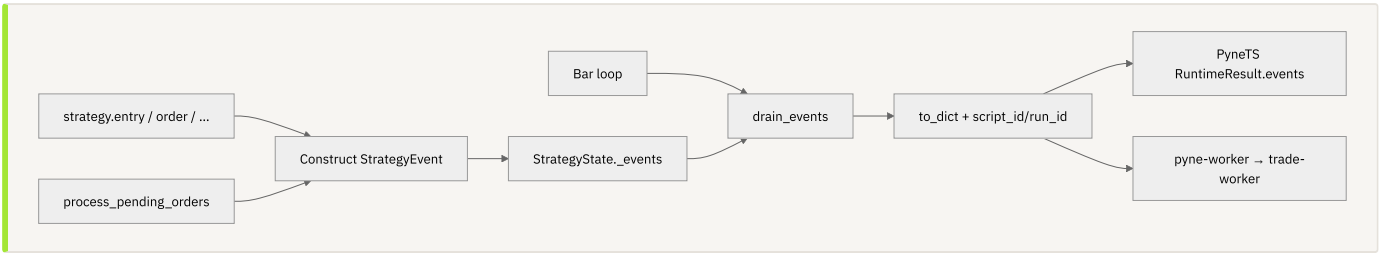
title: "Strategy events" description: "StrategyEvent shape, emission points, parity corpus, and host serialization."

STRATEGY EVENTS

Abstract

A **strategy event** is the immutable, structured record of one `strategy.*` action (or fill side-effect) at a specific bar. Events are the wire format between PYNE's Python evaluator, **PyneTS** (`StrategyEvent` on `RuntimeResult`), and downstream HOOX trade workers (via **pyne-worker**). Changing the dataclass without updating the TS library and tests is a contract break.

Conceptual model



Interface surface

`StrategyEvent` fields

Defined in `src/pynescript/ast/evaluator/events.py`:

Field	Type (logical)	Notes
`kind`	`entry` | `exit` | `close` | `close_all` | `cancel` | `cancel_all` | `order`	Discriminator
`id`	`str | None`	Order / entry id
`direction`	`long` | `short` | `None`	
`qty`	`float | None`	
`order_type`	`market` | `limit` | `stop` | `None`	Stop-limit may appear as composite handling elsewhere
`limit` / `stop`	`float | None`	Prices when relevant
`oca_name`	`str | None`	OCA group
`comment`	`str | None`	Includes system comments (`risk_blocked` , `fill:…`)
`bar_index`	`int`	From context at emit
`bar_time`	`int`	From context
`ohlc`	4-tuple → list in JSON	Bar snapshot
`script_id`	`str`	Host-stamped (source hash)
`run_id`	`str`	Host-stamped run uuid

Frozen dataclass → `to_dict()` always includes keys (JSON nulls for unspecified fields) so parity diffs are stable.

Emission sources

1. **Explicit builtins** — `strategy.entry` , `exit` , `close` , `close_all` , `cancel` , `cancel_all` , `order` push events as they execute.
2. **Broker fills** — `process_pending_orders` emits fill-related `order` / `entry` / `cancel` (OCA) events.
3. **Risk gates** — blocked entries may emit with `comment="risk_blocked"` .

Host collection pattern

```
evaluator.reset_events()
# ... process_pending_orders + visit(tree)
for ev in evaluator._strategy_state.drain_events():
    d = ev.to_dict()
    d["script_id"] = script_id
    d["run_id"] = run_id
    all_events.append(d)
```

`Runtime.run` implements this and returns events in the response envelope.

Compile path

`CompileStrategyBroker` accumulates plain dict events (`__events` in the return mapping) with the same field names for kinds it supports. Exit/trail/OCA/risk halt comments (`risk_blocked`) are on both hosts for the 0.3.4+ surface; remaining metric accessors still trail the interpreter.

Internals

Path	Role
`src/pynescript/ast/evaluator/events.py`	`StrategyEvent`
`src/pynescript/ast/evaluator/builtins/strategy.py`	Emit sites + `drain_events`
`src/pynescript/compiler/strategy_broker.py`	Compile-mode `_emit`
`tests/test_strategy_events.py`	Unit behavior
`tests/test_parity.py` + `tests/fixtures/parity/`	Cross-port oracle
`pynets/src/runtime/interpret.ts` `StrategyEvent`	TS library twin

Parity corpus scripts (`strategy_01_entry_long.pine`, …) regenerate expected JSON via:

```
python tests/fixtures/parity/generate_fixtures.py
```

Tests strip `script_id` / `run_id` before compare.

Invariants & edge cases

1. **Immutability.** Do not mutate events after construction; drain copies and clears the buffer.
2. **Per-bar reset.** `reset_events` / drain prevents cross-bar leakage in tests that share evaluators.
3. **OHLC list vs tuple.** `to_dict` converts to list for JSON round-trips.
4. **Kind vocabulary is closed.** New kinds require TS + fixtures + this doc.
5. **Not alerts.** `alert()` / `alertcondition()` use a separate alerts channel—not `StrategyEvent`.

Worked examples

Expected single entry

Script enters long on a fixed bar; fixture JSON lists one event:

```
{
  "kind": "entry",
  "id": "L",
  "direction": "long",
  "qty": 1.0,
  "order_type": "market",
  "bar_index": 5
}
```

(plus nullables and ohlc—see real fixtures for full keys).

Cancel after OCA

Fill of one OCA sibling yields `cancel` events for others with `comment` reflecting `oca_cancel` / `oca_reduce`.

Failure modes

Symptom	Cause
Empty events with live strategy	Forgot drain; or strategy never called
Parity flake on ids	Comparing `script_id` / `run_id`
TS/Python mismatch	Field rename without dual update
Duplicate events	Not clearing buffer; double `process_pending_orders`

See also

- Strategy builtins
- Strategy broker
- PyneTS
- pyne-worker
- HOOX live trading
- HOOX docs

ALERTS

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title: "Alerts" description: "alert() / alertcondition() engine, host export, and L2 HTTP webhooks (Pro API + pyne-worker)."

ALERTS

Abstract

Pine `alert()` and `alertcondition()` fire into a **host side-channel**, not strategy events. The evaluator records structured firings with TradingView-style frequency rules. Hosts (Pro API and **pyne-worker**) export them on evaluate responses and can **POST** them to HTTP webhooks (roadmap **L2**).

Conceptual model



Layer	Module
Engine	`src/pynescript/ast/evaluator/builtins/alerts.py` (`export_alerts` / `export_alerts_from_evaluator`)
Package host	`src/pynescript/runtime/host.py` (`clear_alerts` per run, `alerts` on the envelope)
Pro API webhook	`backend/alert_forwarder.py` (`backend/runtime.py` is only a Runtime re-export)
Edge host	sibling pyne-worker `alert_engine.py` + `alert_forwarder.py`

Pine surface

`alert(message, freq)`

- Default `freq` : `alert.freq_once_per_bar` (canonical token `once_per_bar`).
- Constants (also bare strings): `alert.freq_once_per_bar` , `alert.freq_once_per_bar_close` , `alert.freq_all` .
- **once_per_bar** — first fire per bar for the same (source, title, message, freq) key.
- **once_per_bar_close** — only when the host marks the bar confirmed (`barstate.isconfirmed` / `islast` ; hosts without barstate treat bars as confirmed).
- **all** — every call.

`alertcondition(condition, title, message)`

- Records each evaluation in `alert_conditions` (debug / UI).
- When `condition` is true, also **fires** a host alert with `source: "alertcondition"` and once-per-bar semantics.

Event shape (`to_dict` / API)

```
{
  "message": "cross up",
  "freq": "once_per_bar",
  "bar_index": 42,
  "time": 1700000000000,
  "title": "optional",
  "source": "alert"
}
```

`source` is `"alert"` or `"alertcondition"` . Hosts stamp `script_id` / `run_id` (and edge cron may add `symbol` / `timeframe`).

Host export

Pro API (`POST /run`)

- Interpret path: full history of firings in `alerts` (and optional `alert_conditions`).
- Compile path: `alerts: []` today (alert side effects are interpret-only).
- See POST /run.

pyne-worker

- `POST /run` returns the same `alerts` array.
- Cron filters to the **last closed bar** before webhook delivery so historical bars do not re-fire every minute.
- Health: `features.alerts`, `features.alert_webhooks`.

L2 webhooks

Both hosts POST JSON to an HTTPS endpoint.

Host	Default URL	Per-request / per-job
Pro API	env `ALERT_WEBHOOK_URL`	body `webhook_url`
pyne-worker	secret/var `ALERT_WEBHOOK_URL`	script/job `webhook_url`

Pro API request flags

Field	Default	Meaning
`webhook_url`	`""`	Override destination
`forward_alerts`	`true`	Skip POST when false
`alert_last_bar`	`true`	Only firings on the last OHLCV bar
`alert_batch`	`true`	One batch body vs one POST per alert

Optional timeout: env `ALERT_WEBHOOK_TIMEOUT` (seconds, default 10).

Batch body

```
{
  "type": "pine_alert_batch",
  "source": "pyne-pro-api",
  "count": 1,
  "content": "cross up",
  "alerts": [
    {
      "type": "pine_alert",
      "source": "pyne-pro-api",
      "message": "cross up",
      "freq": "once_per_bar",
      "alert_source": "alert",
      "bar_index": 42,
      "time": 1700000000000
    }
  ]
}
```

Edge worker uses `"source": "pyne-worker"`. Discord-friendly `content` is set when a message is present.

Response meta

Successful `/run` may include:

```
"alert_forward": {
  "forwarded": 1,
  "failed": 0,
  "filter": "last_bar",
  "url": "https://hooks.example.com/pine",
  "batch": true,
  "count": 1
}
```

Webhook failures never fail the evaluate status; inspect `alert_forward` / `alert_forward_error`.

Worked example

```
curl -sS -X POST http://127.0.0.1:5002/run \
  -H 'Content-Type: application/json' \
  -d '{
    "script": "//@version=6\nindicator(\"a\")\nif bar_index == last_bar_index\n    alert(\"fire\")\
    "mode": "interpret",
    "webhook_url": "https://hooks.example.com/pine",
    "data": [
      {"time": 1, "open": 1, "high": 2, "low": 0.5, "close": 1.5, "volume": 10},
      {"time": 2, "open": 1.5, "high": 2.5, "low": 1.0, "close": 2.0, "volume": 12}
    ]
  }'
```

Invariants

1. **Not strategy events** — trade mesh uses `events[]`; alerts are independent.

2. **Per-run clear** — hosts call `clear_alerts()` so runs do not leak firings.
3. **Last-bar default for webhooks** — full-history still available in the `alerts` response field.
4. **Compile path** — prefer `mode: "interpret"` (or auto fallback) when you need alerts.

See also

- Strategy events
- POST /run
- Pro API usage
- Runtime bridge
- Roadmap (L2 ✅)

OVERVIEW

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---

title: "Compiler overview" description: "Source-to-source compilation: CompilerVisitor, numeric vs object mode, run(time=...), and engine API."

COMPILER OVERVIEW

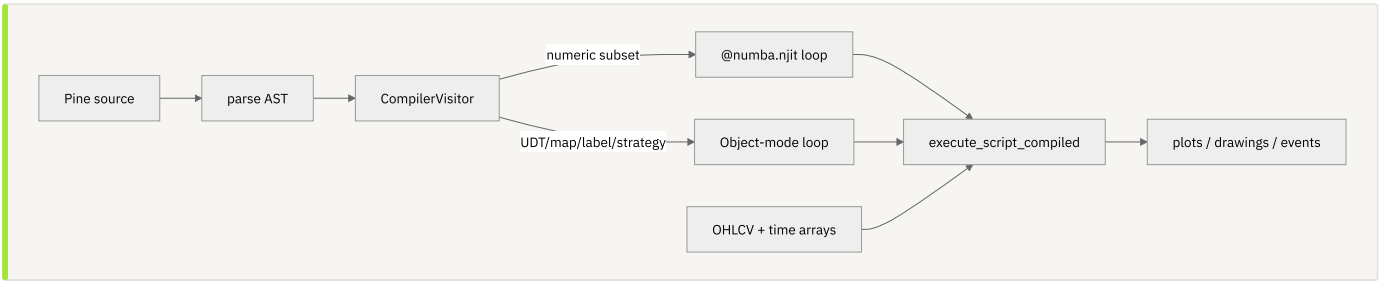
Abstract

The compile path lowers Pine's ASDL AST to **executable Python** that walks bars with an explicit index (`__bar_idx`) over contiguous `numpy` arrays—avoiding per-node visitor overhead. Pure numeric scripts wrap the loop in `@numba.njit`; scripts that touch UDTs, maps, drawings, or strategy select **object mode** (a tight Python/numpy loop, no AST walk). **Numba is required only for numeric mode**; object mode runs with numpy alone. **`once`** (0.4.4) lowers to a persistent `__once_fired_N` flag; UDF-scoped `once` forces object mode.

The public façade is `pynescript.compiler.engine` (`transpile`, `compile_script`, `run_script`), also reachable as `Runtime.run(..., mode="compile")` or `mode="auto"`.

This is an MVP with growing surface—not a full substitute for the interpreter on every script (notably multi-asset `request.*` and full strategy analytics). **0.3.16–0.3.17** recovered a large object-mode residual class: UDF locals, nopython `None`/unicode helpers, matrix handles, UDT returns, statement `switch` assign, drawing/`chart.point` copy, `round()` extras, import stubs as `na`, color type dispatch, and nested UDT method fields. Official TV builtins are **0 MISMATCH** vs interpret (**P1p** closed on that set); optional set0x community corpus is unmeasured, not a missing language surface.

Conceptual model



Generated entry point is always:

```
execute_script_compiled(open_arr, high_arr, low_arr, close_arr, vol_arr, time_arr)
```

Host packing lives in `engine.CompiledScript.run` (and `pynescript.runtime.Runtime._run_compiled`).

Interface surface

```
from pynescript.compiler.engine import transpile, compile_script, run_script, has_numba

src = transpile(pine_text)           # generated Python string
cs = compile_script(pine_text)       # CompiledScript
result = cs.run(open, high, low, close, volume, time=None)
# result: dict of plot title → float64 array, optional __drawings, __events, …
```

API	Role
`transpile`	Parse + visit → source string (inspect/debug)
`compile_script`	`exec` generated code, warm-up call, return `CompiledScript`
`run_script`	One-shot compile+run (prefer cache `CompiledScript` for batches)
`has_numba`	Whether numeric (njit) mode can load
`prewarm_numba_builtins` / `prewarm_scripts`	Host cold-start (H2 warm path)
`compile_cache_stats` / `compile_deploy_config`	Cache + deploy diagnostics
`clear_compile_cache`	In-process LRU only
`clear_disk_compile_cache`	Disk IR / index under cache dir
`clear_numba_function_caches`	Purge Numba `.nbi` / `.nbc` under known dirs

`CompiledScript` fields: `source`, `generated_code`, `execute`, `plot_titles`, `plot_kinds` (per-plot `plot` / `hline` / `fill` tags used to synthesize numeric-mode `__drawings`), `object_mode`, optional `nopython_fallback_reason`.

`CompiledScript.run` signature

```
def run(
    self,
    open_, high, low, close,
    volume=None,   # default: ones
    time=None,     # bar-open Unix ms; optional 6th series
) -> dict[str, Any]:
```

Argument	Behavior
OHLCV	Coerced to contiguous `float64`; lengths must match (`ValueError` otherwise)
`volume=None`	Filled with ones
`time=None`	Synthetic `bar_index * 60_000` ms (unit-test / pure-compile default)
`time=arr`	Real bar-open timestamps; length must match OHLCV

`Runtime._run_compiled` packs OHLCV dicts and **passes real bar times** so calendar/`time` / `timestamp` / `time[n]` stay aligned with interpret. Pure `cs.run(o,h,l,c)` without `time=` is fine for pure-price scripts and fixtures.

When `time` is omitted, the engine synthesizes `np.arange(n) * 60000.0` (1-minute bar opens from epoch). Hosts that care about `year` / `month` / `time[n]` must pass bar-open Unix ms.

Warm compile (product defaults)

Pro API `/run` defaults to **`mode=auto`** (compile when eligible, interpret on fallback with `compile_fallback_reason`). Disk IR cache is **on** (`PYNE_COMPILE_DISK_CACHE=1`); Docker sets `PYNE_COMPILE_CACHE_DIR=/data/compile-cache`. Operators call `POST /compile/prewarm` or `pynescript prewarm` so first interactive latency skips Numba cold JIT. Design notes live in repo `docs/COMPILER_PLAN.md` (not a product page).

`Runtime._compile_eligible` (used by `mode=auto`) **skips compile** when the source contains a top-level `import` or any `request.` token — those stay interpret even though the emitter can lower same-symbol `request.security`. Explicit `mode=compile` still attempts the visitor.

Mode selection

`CompilerVisitor` sets `object_mode = True` when it sees:

- User-defined types / enums
- Map / array / matrix handles that cannot stay float64
- Drawing constructors (`label.new`, `line.new`, …) and assigned `hline` handles
- Strategy usage (object mode + `CompileStrategyBroker`)
- String/color series, library imports, or other non-numeric constructs

Statement-form `hline(price)` / `fill(...)` with a proven-numeric price **stay nopython**: they emit a constant (or all-`nan`) series plus `plot_kinds` metadata; the engine synthesizes `__drawings` after the JIT loop. Mixed scripts that later flip object mode replay those events into the body.

Otherwise **numeric mode** is preferred. Numeric mode **requires Numba** at load/warm-up; missing Numba raises `CompileNumbaRequiredError`. Object mode does **not** need Numba (still star-imports `numba_builtins` for `safe_*` / non-jitted helpers).

On numeric warm-up **TypingError** / nopython failure, the engine re-emits with `force_object_mode=True` and records `nopython_fallback_reason`—so `mode="compile"` can still run without hard-failing on partial numeric coverage.

Numba is **not** a hard requirement for all compile. Object-mode scripts (strategy, drawings, UDT, maps) execute without it. Only a pure-numeric emit that never flips to object mode needs Numba installed.

What numeric mode lowers today

Series assigns, arithmetic/logic, history (`close[n]`), `if` / `for` / `while`, selected `ta.*` (and bare TA aliases), scalar math helpers, `plot`, `input.*` defaults, `var` / `varip` carry—executed under `@numba.njit(cache=False)` on the generated entry (`cache=False` because code is `exec`'d from a string without a stable disk locator). Disk-cached IR modules may rewrite to `cache=True` for cross-process reuse of machine code.

Object mode extras

- UDT instances as field dicts
- Maps as Python dicts
- `__drawings` list of structured events
- Optional `__strategy` broker: pending fills, position, `__events`, equity snapshots

Compile-path limits (notable)

Construct	Compile behavior
`request.security` / bare `security` / `request.security_lower_tf` / `request.seed`	**Same-symbol, simple OHLCV expression only** (chart identity + bare `close` / `high` /... style). Foreign symbol or complex expression → `np.nan` (no inventing chart close as foreign data). Other `request.*` → object-mode `np.nan` stub
Bare builtins vs user series	User-defined series arrays win in `visit_Name`: `ad = ta.cum(...)` / `tr = …` shadow bare `ad` / `tr` formulas so `plot(ad)` is not silently re-bound to the builtin
`hline(...)`	Statement-form + numeric price → **nopython** constant series (`hline`, `hline_2`, ...) + synthesized `__drawings`. Assigned / non-numeric handles → object mode
`fill(..., title=…)`	Statement-form → **nopython** all-`nan` series key (band color in `plot_meta` / drawings). Expression-form / mixed object scripts emit `__drawings` in-loop

Compile-path `request.security` never invents foreign OHLCV from chart close. Foreign tickers and complex expressions (UDFs, `ta.*` inside the third arg, multi-value constructs beyond simple chart series) lower to `np.nan`. Use interpret + a real `data_feed` / `data_provider` for multi-asset work.

Internals

Path	Role
`src/pynescript/compiler/compiler.py`	`CompilerVisitor`, emit numeric/object, security/name/fill/hline lowering
`src/pynescript/compiler/numba_builtins.py`	JIT kernels + object-mode `safe_*`
`src/pynescript/compiler/strategy_broker.py`	Compile broker
`src/pynescript/compiler/engine.py`	Façade, caches, `CompiledScript.run`, Numba cache recovery
`src/pynescript/runtime/host.py`	Package SoT `Runtime._run_compiled` envelope; packs OHLCV + `time=`
`backend/runtime.py`	Compat re-export of `pynescript.runtime.host`
`scripts/compare_interp_compile.py`	Interpret↔compile series parity harness
`tests/test_compiler_numba.py`, `test_compiler_objects.py`, `test_compiler_strategy.py`	Coverage
`docs/COMPILER_PLAN.md`	Design history and remaining work (repo, not hosted)

Data layout

Unlike interpreter `PineSeries` deques:

```
open_arr, high_arr, low_arr, close_arr, vol_arr, time_arr
user_arr = np.full(n_bars, np.nan)                # or dtype=object for UDT
plot_i   = np.full(n_bars, np.nan)
for __bar_idx in range(n_bars):
    user_arr[__bar_idx] = …
    plot_i[__bar_idx] = …
```

- Bare `time` / `time_close` / `last_bar_time` / calendar extractors read `time_arr` (synthetic when omitted).
- `var` lowers to “init only when still na / first write” patterns so carry matches Pine without declaration sets.
- Numeric mode returns a **tuple** of plot arrays (host maps titles); object mode returns a **dict** (plots + `__drawings` / strategy extras).

Caches and corrupt-Numba recovery

Three layers:

1. **In-process source LRU** (sha256 of raw and/or sanitized source) + secondary IR cache keyed by generated-code hash.
2. **Disk IR** (default on): modules under `PYNE_COMPILE_CACHE_DIR` or `$XDG_CACHE_HOME/pynescript/compile`. The source→IR index JSON carries `"v": 9` (`engine._DISK_META_VERSION`). Bump that integer when generated IR semantics change so stale modules are ignored (source hash alone is stable across emitter fixes).
3. **Numba function cache** (`.nbi` / `.nbc`) next to `numba_builtins` and under disk-IR `__pycache__/`.

Truncated/corrupt Numba pickle files raise `EOFError` / `pickle.UnpicklingError` on load. The engine wraps execute/warm paths with **purge + single retry** via `clear_numba_function_caches` (does not fail the script for a bad cache alone). Manual clean slate:

```
from pynescript.compiler import (
    clear_compile_cache,
    clear_disk_compile_cache,
    clear_numba_function_caches,
)
clear_compile_cache()
clear_disk_compile_cache()
clear_numba_function_caches()
# shell: rm -rf ~/.cache/pynescript/compile \
#   src/pynescript/compiler/__pycache__/numba_builtins*.nb*
```

Clear disk + Numba caches after compiler emitter or kernel edits if results look stale.

Invariants & edge cases

1. **Same AST front-end.** No second parser—compile bugs are lowering bugs.
2. **OHLCV (+ time when provided) length equality** enforced in `CompiledScript.run`.
3. **Warm-up** ignores exceptions on a dummy 16-bar series (JIT or first-run); non-nopython failures may surface on the first real run.
4. **Result normalization** converts Numba typed maps to plain dicts; `__drawings` / `__events` stay Python lists.
5. **User series shadow bare builtins** in `visit_Name` (`ad`, `tr`, `n`, ...) once `*_arr` is allocated.
6. **`request.security` is not multi-asset on compile.** Only chart-symbol simple OHLCV passthrough; everything else is `na`.
7. **Interpret still owns the full surface.** Exotic builtins, real multi-symbol feeds, and rich strategy metrics remain interpret-first until lowered.

Worked example

Pine:

```
//@version=6
indicator("c")
s = ta.sma(close, 14)
plot(s, "sma")
```

Generated shape (illustrative; numeric mode returns a tuple of series):

```
@numba.njit(cache=False)
def execute_script_compiled(open_arr, high_arr, low_arr, close_arr, vol_arr, time_arr):
    n_bars = len(close_arr)
    s_arr = np.full(n_bars, np.nan)
    plot_0 = np.full(n_bars, np.nan)
    for __bar_idx in range(n_bars):
        s_arr[__bar_idx] = numba_sma(close_arr, 14, __bar_idx)
        plot_0[__bar_idx] = s_arr[__bar_idx]
    return (plot_0,)
# Host packs titles → {'sma': plot_0}
```

Interpret ↔ compile parity

Full contract (tolerances, buckets, intentional diffs): Interpret ↔ compile parity.

```
# From repo root — nan-aware allclose on common series keys
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --glob 'ta_*.pine' --bars 200
python scripts/compare_interp_compile.py --files tests/fixtures/parity/pine/strategy_01_entry_long.
python scripts/compare_interp_compile.py --ignore-hline-keys --ignore-fill-keys --strict-keys
```

Writes `.cache/interp_compile_parity.json`. First-party `hline` / `fill` / `bgcolor` / `plotshape` / `barcolor` / `plotarrow` / `plotbar` / `plotcandle` keys match interpret ↔ compile (0.3.12); `drawings` is geometry-only. Ignore flags are for leftover corpus noise.

Failure modes

Symptom	Cause
`numba is required for numeric compile mode` (`CompileNumbaRequiredError`)	Install numba **or** force/land in object mode (drawings/UDT/strategy already do)
Empty generated code	Visitor returned blank—unsupported top-level shape (`CompileEmitError`)
Missing `execute_script_compiled`	Emit bug or failed `exec` (`CompileLoadError`)
`EOFError` / `UnpicklingError` during JIT load	Corrupt Numba `.nb*` —engine purges and retries; if stuck, call `clear_numba_function_caches`
Silent semantic drift	Kernel seed differences—run `scripts/compare_interp_compile.py`
All-`nan` from `request.security`	Foreign symbol or non-simple expression on compile path (by design)
Wrong values for `ad` / `tr` after assignment	Fixed by user-series shadowing; if stale, clear compile/Numba caches
Strategy metrics missing	Use interpret or inspect `__events` / broker fields only
Calendar/`time` off by hours	Host omitted `time=`; synthetic 1m bar opens used

Remaining work (summary)

Expand Numba `ta.*` surface, richer drawings, nested UDT methods, broader `request.*` (still interpret-first for multi-asset; `mode=auto` already skips any `request.` source), and grow systematic interpret↔compile parity coverage per lowered construct. In-process + disk IR cache (`_DISK_META_VERSION = 9`) and product prewarm are landed. See repo `docs/COMPILER_PLAN.md` for the living checklist.

See also

- Numba path
- Strategy broker
- Interpret ↔ compile parity
- Runtime hub
- request.* / input.* (interpret)
- Numerical validation

NUMBA

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title: "Numba path" description: "JIT bar loops, numba_builtins kernels, NA-as-NaN, and performance characteristics."

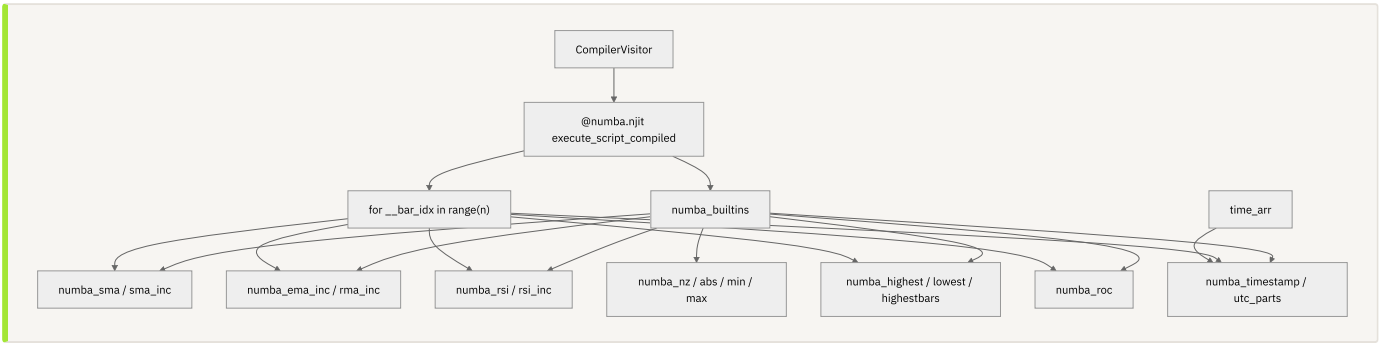
NUMBA PATH

Abstract

Numeric compile mode emits a single `@numba.njit` function whose body is the script's bar loop. Pine builtins that cannot cross the JIT boundary are replaced by **hand-written kernels** in `numba_builtins.py` that take full arrays plus the current index (or a small float state vector for `*_inc` forms). The result is large speedups on multi-thousand-bar series after a one-time JIT warm-up—at the cost of a restricted language subset and `np.nan` as the missing-value sentinel.

Kernel contracts below match interpret `ta.*` / calendar helpers where parity was fixed; intentional deviations should be documented on the kernel docstring and here.

Conceptual model



Interface surface

Kernels are defined with `@numba.njit(cache=True)` in `numba_builtins.py`. Generated entry points use `@numba.njit(cache=False)` in the in-memory IR string; when the engine writes a **disk IR** module it rewrites that decorator to `cache=True` so Numba can store machine code next to a real file path (see Cache).

Generated code imports `from pynescript.compiler.numba_builtins import *` so call sites are bare names. Prefer `*_inc` when the visitor allocates fixed state (`__ema0_st` , ...).

Kernel table (interpret-parity highlights)

Kernel	Semantics
`numba_sma(arr, period, i)`	Mean of last `period` bars ending at `i`; `nan` if warm-up or any window `nan`
`numba_ema` / `numba_ema_inc`	SMA seed, then EMA (`α = 2/(period+1)`). **Inc seed is NaN-safe**: while the accumulator is still `nan` and `j >= period-1`, try SMA over `arr[j-period+1 : j+1]` only when every sample is finite. Nested EMA / DEMA no longer stuck all-NaN after leading NaNs
`numba_rma_inc(arr, period, i, st)`	Wilder RMA with the **same NaN-window-safe SMA seed** as `numba_ema_inc`. After seed, NaN inputs hold the previous RMA. Required for `rma(tr)` / expanded ADX so compile is not all-NaN or ~0 after warm-up
`numba_rsi` / `numba_rsi_inc`	**Wilder RSI**, not rolling simple-window RSI. SMA of first `period` deltas (bars `1..period`), then RMA of gain/loss (`α = 1/period`). First valid bar is **`i == period`** (`period` deltas need `period+1` prices). `avg_loss == 0` → `100.0`
`numba_highest` / `numba_lowest`	Window extrema; full window required (`i >= period-1`); NaN samples skipped
`numba_highestbars` / `numba_lowestbars` (+ `_inc`)	TV **negative** bars-back offsets: `0` if current is extreme, `-1` one bar ago, ..., `-(length-1)`. Returns **`-1.0`** when window not full, invalid length, or all-NaN. **Oldest** extreme wins ties (strict `>` / `<`)
`numba_roc(arr, length, i)`	`100 * (src - src[length]) / src[length]` at bar `i`; `nan` if `i < length`, baseline 0/NaN, or src NaN
`numba_timestamp(y, m, d, h=0, mi=0, s=0)`	UTC epoch **ms** from calendar components; month/day overflow normalized (e.g. `month=0` → prior Dec) for TTM-style windows
`numba_utc_parts(ms)`	`(year, month, dayofmonth, hour, minute, second, dayofweek)` floats; Pine `dayofweek` is `1=Sun … 7=Sat`
`numba_nz(val, replacement)`	Replace `nan`
`numba_abs` / `numba_min` / `numba_max`	Scalar helpers

Full inventory lives in `src/pynescript/compiler/numba_builtins.py` (njit batch + `*_inc` + pure-Python object-mode helpers).

Calendar / `time_arr`

Bar open time is the host `time_arr` series (synthetic when the runner omits timestamps).

Pine form	Lowering
bare `year` / `month` / `dayofmonth` / `hour` / `minute` / `second` / `dayofweek`	`numba_utc_parts(time_arr[__bar_idx])[idx]`
`year(t)` ... (optional time arg)	`numba_utc_parts(src)[idx]` with `src` defaulting to `time_arr[__bar_idx]`
`timestamp(y, m, d[, h, mi, s])`	`numba_timestamp(...)` (leading timezone string args dropped → UTC stub)
`time` / `time_close` (approx)	`time_arr[__bar_idx]` / `+ 59999` ms

History access

`close[n]` lowers to array indexing at `__bar_idx - n` with bounds → `nan` (parallel to interpreter `None`, different sentinel).

Control flow

`if` / `for` / `while` / `once` emit Python control flow **inside** the njit function—supported Numba subset only (no heap objects, no Python lists of mixed types). `once` lowers to a persistent `__once_fired_N` bool set when the body runs (compile is historical: `barstate.isconfirmed` is always true). A `once` inside a UDF forces object mode so the flag is module-level.

Cache

Three layers interact:

Layer	Location / key	Role
Source LRU	in-process, sha256 of source	`CompiledScript` reuse
IR LRU	in-process, sha256 of generated Python	share warm callable across comment-only variants
Disk IR	`PYNE_COMPILE_CACHE_DIR` or `$XDG_CACHE_HOME/pynescript/compile`	persist generated modules; disable with `PYNE_COMPILE_DISK_CACHE=0`. Index JSON `"v"` must equal `engine._DISK_META_VERSION` (**9**)

Numba function cache (`.nbi` / `.nbc`):

- Hand-written kernels: `@njit(cache=True)` → artifacts under `…/pynescript/compiler/__pycache__/`.
- Generated entry: IR is emitted with `cache=False` (safe for `exec` from `<string>`). Disk write rewrites `@numba.njit(cache=False)` → `cache=True` so Numba's file locator can cache machine code under the disk IR `__pycache__/`.
- Truncated/corrupt pickle loads raise `EOFError` / `UnpicklingError`. The engine purges known `.nbi` / `.nbc` via `clear_numba_function_caches()` and recompiles once instead of failing the script.

```
from pynescript.compiler import clear_numba_function_caches, clear_disk_compile_cache
clear_numba_function_caches()
clear_disk_compile_cache()
```

Internals

Path	Role
`src/pynescript/compiler/numba_builtins.py`	Kernels (`cache=True`)
`src/pynescript/compiler/compiler.py`	`_emit_numeric_mode`, call lowering, `time_arr` calendar
`src/pynescript/compiler/engine.py`	Disk IR rewrite, Numba cache recovery, prewarm
`tests/test_compiler_numba.py`	Correctness / parity

Expanding coverage is primarily **new kernels + visitor call mapping**, not changes to the bar-loop skeleton.

Invariants & edge cases

1. **No Python objects in numeric mode.** Strings, UDTs, maps, drawings force object mode before njit is applied.
2. **Warm-up cost.** First `compile_script` invokes a dummy run; production servers should cache `CompiledScript` and prewarm (`PYNE_COMPILE_PREWARM` default **on**, `POST /compile/prewarm`).
3. **EMA/RMA seed** uses sliding all-finite SMA windows—not "seed once at bar `period-1` only if that fixed slice is clean." Leading NaN on `tr` / nested EMA sources must eventually produce values.
4. **RSI is Wilder**, first valid at `i == period` . Do not compare against a simple N-bar average-gain RSI.
5. **highestbars/lowestbars** return negative offsets and `-1` on incomplete windows; ties keep the oldest bar.
6. **Disk entry cache** requires the rewrite to a real file path; pure `exec` of `cache=False` IR does not populate Numba's on-disk function cache for the entry function.
7. **Anaconda static libpython.** Packaging notes in AGENTS/build docs apply when shipping JIT-heavy binaries.

Worked examples

Benchmark mental model

```
interpret:  O(bars × ast_nodes × python_dispatch)
numeric:    O(bars × lowered_ops) in machine code after JIT
```

For simple SMA scripts, internal notes cite order-of-magnitude speedups versus list-based interpretation on long series (see repo `docs/COMPILER_PLAN.md` qualitative claims).

Using from Pro API

```
Runtime(...).run(source, ohlcv, mode="compile")
```

Converts bar dicts → arrays (including `time=` bar-open ms), compiles, reshapes plots into the standard envelope, flags `object_mode` when applicable. `generated_code` is omitted unless `PYNESCRIPT_RETURN_GENERATED_CODE=1` .

Failure modes

Symptom	Cause
TypingError from Numba	Unsupported construct leaked into numeric emit
All-`nan` plots	Period longer than series; history index bugs; or (fixed) NaN-poisoned EMA/RMA seed on nested sources
Nested DEMA / `rma(tr)` all-NaN	Pre-fix seed; current kernels sliding-window all-finite SMA seed
RSI diverges vs interpret	Expect Wilder seed/RMA, not rolling window RSI
`highestbars` sign / magnitude off	TV negative offsets; incomplete window → `-1`
`EOFError` / UnpicklingError at load	Corrupt `.nbi` / `.nbc` — engine purges and retries; manual `clear_numba_function_caches()`
ImportError numba	Environment missing dependency
Object mode unexpectedly	Visitor detected drawing/UDT/map—inspect `visitor.object_mode`

Performance notes

- Prefer **one compile, many runs** with different OHLCV.
- Keep scripts in the numeric subset for realtime ticks.
- Host prewarm of `numba_builtins` + common scripts shifts cold JIT off the first user request.
- Object mode still wins over AST walking for UDT-heavy scripts but will not match peak njit throughput.

See also

- Compiler overview
- Technical builtins (interpret)
- Series & history

STRATEGY BROKER

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title: "Compile strategy broker" description: "CompileStrategyBroker: pending orders, OHLC fills, OCA, commission, and event export."

COMPILE STRATEGY BROKER

Abstract

When `CompilerVisitor` detects strategy usage, object-mode emission instantiates **`CompileStrategyBroker`**—a lightweight broker aligned with the interpreter's fill-before-script cadence, but designed for generated loops (no AST, no full `StrategyState` metric surface). It supports market entry/close and pending limit/stop/stop-limit orders with per-bar OHLC triggers, OCA cancel/reduce, commission, and slippage.

Conceptual model



Syntax error in text

mermaid version 11.17.2

Interface surface

Construction (from `strategy()` kwargs when lowered)

```
CompileStrategyBroker(
    initial_capital=100_000.0,
    commission_value=0.0,
    commission_type="percent",  # percent | cash_per_order | cash_per_contract
    slippage_ticks=0,
    mintick=0.01,
    pyramiding=0,
    default_qty_type="fixed",
    default_qty_value=1.0,
    avg_price_model="stock",  # stock | futures | inverse (pynescript extension)
    leverage=1.0,             # futures UI: buying power / margin divisor
)
```

`avg_price_model` is lowered from `strategy(..., avg_price_model=...)`. The compile broker is a **single net lot**: partial reduce already keeps `position_avg_price` sticky (futures-like). Multi-leg stock reweight on partial close is interpret-first until the compile path grows an open-trade list.

`leverage` scales percent/cash default qty (`margin × leverage / price`) and free `cash` (equity – notional/leverage).

Bar context

```
# Preferred generated hot path (one call; skips pending walk when empty)
broker.begin_bar(bar_index, open_, high, low, close, bar_time=…)
# Equivalent two-step still exists:
broker.set_bar(bar_index, bar_time, mark, open_=…, high=…, low=…, close=…)
broker.process_pending_orders(open_=…, high=…, low=…, close=…)
```

Orders

Method	Role
`entry(...)`	Market or pending entry; reverse flattens opposite
`close` / `close_all`	Reduce/flat with optional price
Pending book	`PendingOrder` with type, direction, qty, limit/stop, OCA, `max_fill_per_bar` , `is_entry`

Fill logic (`_trigger_price`)

Type	Long trigger	Short trigger	Fill price idea
market	—	—	close (immediate path)
limit	`low <= limit`	`high >= limit`	gap-aware vs open
stop	`high >= stop`	`low <= stop`	gap-aware vs open
stop-limit	stop traded and limit available	symmetric	limit price

Partial fills: `max_fill_per_bar` caps quantity per bar; residual stays pending.

OCA

After a fill, siblings in the same `oca_name`:

- **`cancel`** — remove sibling, emit `cancel`
- **`reduce`** — decrease sibling qty by fill size
- **`none`** — no interaction

Economics

- **Slippage:** `± slippage_ticks * mintick` by direction on fills.
- **Commission:** percent of notional, cash per order, or cash per contract.
- **Position:** signed `position_size`, `position_avg_price`, `netprofit`, `equity` helpers for return payload.

Events

`_emit` appends dicts with keys compatible with strategy event serialization (`kind`, `id`, `direction`, `qty`, `order_type`, `limit`, `stop`, `oca_name`, `comment`, `bar_index`, `bar_time`, `ohlc`). Object-mode return includes `'__events': __strategy.to_events()` plus position/netprofit/equity snapshots.

Internals

Path	Role
`src/pynescript/compiler/strategy_broker.py`	Broker implementation
`src/pynescript/compiler/compiler.py`	Emits `__strategy` wiring in object mode
`src/pynescript/ast/evaluator/builtins/strategy.py`	Interpreter counterpart (richer)
`tests/test_compiler_strategy.py`	Compile strategy coverage

NA helpers treat `None`, blank/`na` strings, and float NaN as missing prices when classifying order types.

Invariants & edge cases

1. **Same order as interpreter:** `begin_bar` (or `set_bar` → `process_pending_orders`) → script body.
2. **Closed-trade list is partial.** The compile broker now keeps `closed_trade_records` and some `closedtrades_*` accessors (profit, comments, max drawdown/runup). Prefer interpret for the full `strategy.closedtrades.*` / `opentrades.*` metric surface.
3. **Direction normalization** accepts `strategy.long`, `long`, `1`, `buy`, etc.

4. **Reverse on opposite entry** via `close_all(comment="reverse")` then open.
5. **Event dicts are mutable lists**—fine for run output; not frozen dataclasses.

Worked examples

Generated prologue (illustrative)

```
__strategy = CompileStrategyBroker(initial_capital=100000.0)
for __bar_idx in range(n_bars):
    __strategy.begin_bar(__bar_idx,
        float(open_arr[__bar_idx]), float(high_arr[__bar_idx]),
        float(low_arr[__bar_idx]), float(close_arr[__bar_idx]),
        bar_time=int(time_arr[__bar_idx]))
    # lowered strategy.entry / close calls
return {..., '__events': __strategy.to_events(),
        '__position_size': __strategy.position_size,
        '__netprofit': __strategy.netprofit,
        '__equity': __strategy.equity}
```

Limit buy pending

```
strategy.entry("L", strategy.long, limit=low - syminfo.mintick)
```

Object-mode lowers to a pending limit; subsequent bars fill when `low` trades through.

Failure modes

Symptom	Cause
No fills	Triggers never met; or market path not invoked
OCA sibling remains	`oca_type` none / name mismatch
PnL vs interpret drift	Commission/slippage/mintick defaults differ
Missing events in API envelope	Host only maps plots—not `__events`

See also

- Strategy builtins (interpret)
- Events
- Compiler overview
- Numba path

PARITY

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title: "Interpret ↔ compile parity" description: "Plot-series parity between Runtime interpret and compile modes: harness, tolerances, and known sentinel differences."

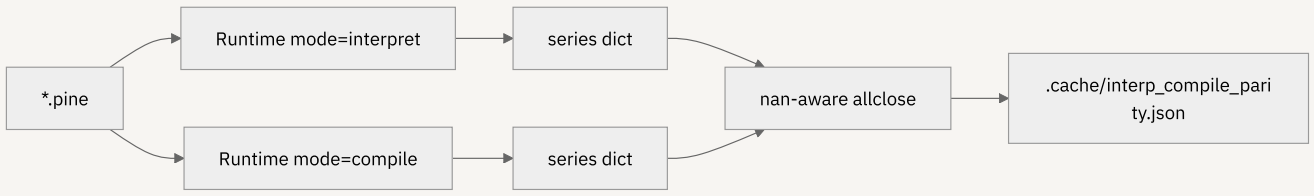
INTERPRET ↔ COMPILE PARITY

Abstract

PYNE ships two bar engines that should agree on **numeric plot series** for scripts in the compile surface: the AST interpreter and the generated Numba/object-mode loop. This page is the short contract for that agreement—how to measure it, what “equal” means, and where intentional differences live (`None` vs `nan` , host `time_arr` , structural keys).

Strategy **event** parity (Python vs TypeScript worker) is a separate oracle (`tests/test_parity.py` + fixtures). Here the focus is **plot values** on one host: `pynescript.runtime.Runtime` .

Conceptual model



Both modes receive the **same** synthetic (or host) OHLCV. Compile additionally builds `time_arr` from bar `time` fields so `time` / calendar expressions line up with the interpret `time` series.

Interface surface

Corpus harness

```
# From repo root — first-party fixtures by default
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --files tests/fixtures/parity/pine/strategy_01_entry_long.
```

Flag / output	Role
Default inputs	`tests/fixtures/parity/pine/*.pine` (first-party)
Tolerances	`rtol=1e-5`, `atol=1e-6` (nan-aware)
Report	`.cache/interp_compile_parity.json`
Exit 0	No value/`nan` mismatches on common series keys

Buckets: `OK`, `fill_background_only`, `both_error_same`, `expected_error`, `both_error`, `MISMATCH`, `interp_error`, `compile_error`, `structural_only`. Matched errors on both backends (`both_error_same`) and intentional demo failures (`expected_error`, e.g. auto-fib / pivot-depth) count as success unless `--strict-errors`. First-party `hline` / `fill` / `bgcolor` / `plotshape` / `barcolor` / `plotarrow` / `plotbar` / `plotcandle` keys match interpret ↔ compile (0.3.12); `drawings` is geometry-only. `--ignore-hline-keys` / `--ignore-fill-keys` remain optional for leftover **corpus** key noise; `--strict-keys` fails on any one-sided key.

Focused unit tests

Test	Scope
`tests/test_dividend_yield_parity.py`	Host-series copy-on-assign + time history (interpret/compile)
`tests/test_compiler_numba.py`	Kernel-level numeric correctness
`tests/test_parity.py`	Strategy **event** fixtures (not plot allclose)

Product path

Pro API `/run` defaults to `mode=auto`: compile when `_compile_eligible` passes (no top-level `import`, no `request.` token), otherwise interpret with `compile_fallback_reason`. Parity harness always forces explicit `interpret` vs `compile` so fallbacks do not hide drift.

Official TradingView® builtins (0.6.2): interpret UDF series-parameter history (`s[1]` per call site) closed the remaining P1p plot MISMATCH tail on that set — **0 MISMATCH** vs compile. Optional set0x community corpus is unmeasured, not a missing language surface.

What must match

1. **Common plot keys** — float series after `None` ↔ `nan` normalization.
2. **Warm-up `na`** — leading missing samples align (interpret `None` / compile `nan`).
3. **`time` / bar-open ms** — when OHLCV includes `time`, or both use the synthetic `i * 60_000` fallback.
4. **Structured errors** — scripts that `runtime.error` (e.g. insufficient pivots) should fail on both with messages that normalize equal (`both_error_same`).

Intentional / soft differences

Topic	Policy
Sentinels	Interpret: `None` ; Numba: `np.nan` . Compare with nan-aware equality, not `==` .
Hline / fill / bgcolor keys	May appear on one mode only; ignore with harness flags unless you need strict key sets.
Unresolved imports / stubs	Non-numeric plot cells serialize as `na` (null)—not string stubs.
Outside compile surface	UDT-heavy / full `request.*` / rich strategy analytics stay interpret-first; `auto` falls back.
Disk / Numba caches	Stale IR after kernel edits can fake mismatches—clear compile + Numba caches (see harness docstring).

Internals

Path	Role
`scripts/compare_interp_compile.py`	End-to-end series compare
`tests/test_interp_compile_parity.py`	Always-on smoke + harness flag tests
`src/pynescript/runtime/host.py`	`_run_compiled` , OHLCV + `time_arr` packing
`backend/runtime.py`	Compat re-export of the package host
`src/pynescript/compiler/engine.py`	Compile façade + result normalize
`src/pynescript/compiler/numba_builtins.py`	JIT kernels under test

Worked example

```
python scripts/compare_interp_compile.py \
  --files tests/fixtures/parity/pine/strategy_01_entry_long.pine \
  --bars 500
# → OK  (or MISMATCH with first differing bar/key in the JSON report)
```

Failure modes

Symptom	Likely cause
Systematic offset after warm-up	EMA/RSI seed or window clamp drift in a kernel
Only-compile empty plots	Script still object-mode / plot not lowered; or cache corruption
`time` diverges	Host omitted times on one path only; check `time_arr` packing
Structural-only keys	hline/fill/background naming — use ignore flags or align collectors

See also

- Compiler overview
- Numba path
- Series & history
- Runtime hub
- Numerical validation
- PyneTS parity — TypeScript vs Python plots